



Research Paper

Numerical study on heat transfer and flow characteristics of symmetric Tesla-type microchannel heat sinks

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ABSTRACT

This study numerically investigates the thermal performance of multi-stage Tesla valve microchannels (TVM) and a symmetric variant (SYM-TVM) using single-phase deionized water, with mass flow rates ranging from 0.5459 to 1.6377 g/s. Both designs, particularly under reverse flow, exhibit lower peak temperatures and reduced temperature gradients compared to forward flow. The SYM-TVM, characterized by its increased vortices and bifurcations, periodically disrupts and redevelops the thermal boundary layer, enhancing heat transfer efficiency. The TVM exhibits a significantly higher pressure drop than the SYM-TVM across all flow conditions, with the reverse flow in TVM reaching up to 2.48 times that of forward flow and exceeding SYM-TVM, especially at a peak flow rate of 1.6377 g/s. The performance evaluation criteria (PEC) and thermal resistance criteria (PECTR) demonstrate the superior performance of SYM-TVM, with a reduced connection angle between the trunk and helix regions enhancing thermal efficiency. Furthermore, the SYM-TVM-Reverse structure with a 30° interconnection angle demonstrates superior performance compared to the conventional rectangular microchannel (RM), achieving a Nusselt number (Nu) that is five times higher than that of the RM and an overall performance up to 2.59 times greater. These results indicate the SYM-TVM-Reverse's high heat transfer efficiency and its capacity to effectively balance thermo-hydraulic performance, even with increased power dissipation.

1. Introduction

The rapid advancement and integration of electronic devices have led to significant heat production, posing challenges in thermal management. Over 55 % of electrical equipment failures are attributed to elevated temperatures [1], emphasizing the need for efficient heat dissipation techniques [2]. The introduction of microchannel heat sink (MCHS) water cooling by Tuckerman and Pease marked a pivotal development, maintaining pressure drops below 185 kPa and achieving heat flows nearing 790 W/cm² within a 10 mm × 10 mm chip area [3]. Microchannels are categorized based on hydraulic diameter (D_h): macro-channels ($D_h \geq 3$ mm), mini-channels ($200 \mu\text{m} \leq D_h < 3$ mm), and micro-channels ($D_h \leq 200 \mu\text{m}$) [4].

Various cooling fluids can be employed in MCHS with distinct properties, including water, ethylene glycol, refrigerants, and nanofluids [5]. With high thermal conductivity (0.6 W/m·K) and specific heat capacity (4.18 J/g·K), water is commonly used despite its freezing and boiling point limitations. Ethylene glycol offers better thermal stability but lower thermal conductivity (0.258 W/m·K). Refrigerants are

suitable for significant cooling via phase change, while nanofluids by dispersing nanoparticles into base fluids show promising enhancements. For instance, CuO nanofluids can boost heat transfer coefficients by up to 26.5 % compared to pure water [6].

Enhancing MCHS thermal efficiency by designing microchannels is a research focus, where different microchannel flow paths, such as sinusoidal [7], serpentine [8], sawtooth [9], and convergent-divergent channels [10], have been explored. Other designs include double-layered [11], manifold configurations [12], bionic fractal [13], and wavy paths [14]. Sajid et al [7] reported a 28.75 % increase in the Nusselt number and a wall temperature reduction to 33.86 °C using sinusoidal minichannel heat sinks with MgO–water nanofluid. Rahimi et al [15] demonstrated the efficiency of serpentine microchannels over straight or parallel ones. Zhou et al [16] developed models for a sawtooth microchannel, achieving Nusselt number deviations under 18 % and friction coefficient discrepancies below 10 %. Lin et al [17] optimized double-layer MCHS designs, leading to significant reductions in thermal resistance. Tang et al [18] found that optimal manifold and channel designs could reduce thermal resistance by 19.18 %, enhancing temperature uniformity.

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Nomenclature		δT	Temperature difference, K
A	Area, m^2	u_{in}	Inlet velocity, m/s
c_p	Specific heat capacity, J/(kg·K)	u_{ave}	Average velocity, m/s
f	Average Darcy friction factor	ρ	Density, kg/m^3
G	Mass flow rate, g/s	μ	Dynamic viscosity, Pa·s
D_h	Hydraulic diameter, m	θ	Interconnection angle for trunk and helix regions
H	Height, mm	Subscripts	
W	Width, mm	0	Original state
h	Average heat transfer coefficient, $W/(m^2 \cdot K)$	ave	Average
k	Thermal conductivity, $W/(m \cdot K)$	ch	Microchannel
L	Length, mm	h	Hydraulic
Nu	Average Nusselt number	tot	Total
p	Pressure, Pa	sf	Interface between liquid and solid
ΔP	Pressure drop, Pa	f	Fluid
Φ	Pumping power, W	s	Solid domain
PEC	Performance evaluation criterion	in	Microchannel inlet
$PECTR$	Performance evaluation criteria of thermal resistance	out	Microchannel outlet
q	Heat flux, W/m^2	max	Maximum
R_t	Thermal resistance of the maximum temperature, $K \cdot m^2 W^{-1}$	min	Minimum
Re	Reynolds number	x	x-direction
T	Temperature, K	y	y-direction
T_{ave}	Average temperature at the bottom of solid domain, K	z	z-direction

Further advancements include manifold microchannel designs which have been shown to outperform conventional heat sinks. For example, Zhou et al [19] constructed a manifold microchannel heat sink that improved the heat transfer coefficient by 87 %. Yan et al [20] utilized genetic algorithms to optimize Y-shaped fractal networks, achieving a thermal resistance of 0.121 K/W. Sui et al [21] presented that corrugated microchannels, due to induced Dean vortices, enhanced coolant mixing and heat transfer efficiency, especially in laminar flow regions.

Integrating structures, such as vortex generators [22], cavities [23], fins [24], and porous architectures [25] into flow channels disrupts conventional boundary layer formations, enhancing heat transfer and reducing thermal gradients [26]. Zhang et al [27] optimized vortex generator configurations, achieving a 23.6 % improvement in the Nusselt number. Khodabandeh et al [28] introduced rectangular fins and sinusoidal cavities, enhancing fluid dynamics and minimizing friction. Yin et al [29] highlighted the superiority of porous copper microchannel heat sinks over solid counterparts.

Tesla-inspired microchannels, characterized by bifurcation-convergence topologies, also show wide prospect. Sun et al [30] integrated sector and diamond-shaped enhancements into Tesla valve-like microchannel heat sinks, achieving significant increases in Nusselt numbers and Darcy friction factors. Rui et al [31] compared MCTV and MCSB designs, noting a 45 % greater heat transfer coefficient and a 25 % lower pressure drop in MCSB. Li et al [32] engineered a 3D cold plate based on the Tesla valve for SiC power modules, achieving a stable temperature variation of only $\pm 5^\circ C$ across each diode.

Other studies have focused on the unique fluid dynamics in Tesla-inspired channels. For instance, the synergy of structural and fluid dynamic principles results in enhanced heat transfer capabilities. Tesla-inspired designs, like those examined by Qian et al [33], Porwal et al [34], Gong et al [35], and Han et al [36], highlight the effectiveness of bifurcation-convergence topologies in ensuring thorough fluid mixing and improving thermal control efficiency. The channel configurations, working fluids, material properties, microchannel dimensions, heat flux and flow rates used in the reviewed studies are presented in Table 1.

This study conducts a comparative analysis of fluid and heat transfer dynamics in advanced Tesla valve-configured microchannels. Focusing on multi-stage Tesla valve microchannel models (TVM) and

symmetrically structured variants (SYM TVM), the performance parameters, including peak heat source surface temperature, temperature distribution variation, performance evaluation criteria (PEC), and turbulent kinetic energy, are evaluated. The influence of the interconnection angle θ between the trunk and helix regions on the thermal efficiency and fluid behavior of the SYMTVM under reverse flow conditions is also examined. To further demonstrate the practical implications, the thermohydraulic performance of the optimized configuration, specifically the SYMTVM-Reverse with a 30° interconnection angle, is compared with that of a conventional rectangular microchannel heat sink (RM), underscoring the achieved significant enhancements.

2. Methodology

2.1. Physical model

Motivated by Nikola Tesla's 1920 innovation of the Tesla Valve [37], our research introduces a novel multi-stage Tesla Valve Microchannel (TVM) heat sink, as illustrated in Fig. 1a, and it is further designed as Symmetrical Tesla Valve Microchannel Heat Sink (SYM TVM), as shown in Fig. 1d. The typical feature of microchannel heat sinks is the symmetrical arrangement between adjacent microchannels. To balance computational efficiency with accuracy in CFD simulations, a single microchannel as the computational domain and applied symmetrical boundary conditions was selected, as shown in Fig. 1b and 1e. This approach significantly reduces computational cost by focusing on a representative section of the system while still capturing the essential flow and thermal dynamics. The use of symmetrical boundary conditions is justified by our model, which meets periodicity and uniform inlet conditions, ensuring that the behavior across adjacent channels is identical. This method allows to achieve accurate results without the need for simulating the entire microchannel array, making it an effective strategy for optimizing computational resources in preliminary design studies [38,39,40]. Both heat sinks are of the same structural dimensions, with the length $L_x = 100$ mm, the width $L_y = 50$ mm and the height $H = 2$ mm. The computational domain widths for both the TVM and SYMTVM configurations are set at $W = 3.54$ mm, with each having a fluid domain height of $H_{ch} = 1.5$ mm. It is important to note that the

Table1
Important parameters and results of selected studies.

References	Channelconfiguration	Working Fluid	Microchannel			Heat Flux and Flow Rate	Comments
			Material	Width	Height		
Sajid et al [7]	Sinusoidal minichannels	MgO/Water nanofluid	Aluminum	1 mm	50 mm	Heat flux: 25 W, 35 W, 45 W; Flow rate: 0.5–2.0 L/min	28.75 % increase in Nusselt number was achieved, lowest wall temperature at 33.86 °C with 0.01 vol % nanofluid.
Rahimi et al [15]	Serpentine microchannels	Water	PMMA	800 μ m	N/A	Flow rate: 4–151 mL/min; Re: 50–2000	CFD and ANN models predicted friction factor with high accuracy; optimized designs showed reduced pressure drop.
Zhou et al [16]	Sawtooth microchannel	Water	316L Stainless Steel	1.22 mm	2.0 mm	Heat flux: 29.6–1871.1 kW/m ² , Re: 100–4300	Maximum error in <i>Nu</i> correlation was 18 %, and friction factor error is 10 %. Improved design insights for optimizing heat transfer and flow resistance.
Lin et al [17]	Double-layer microchannels	Water	Silicon	200–400 μ m	50–150 μ m	Heat flux: 100 W/cm ² , Flow rate: 200–500 mL/min	Optimized double-layer microchannel design reduced thermal resistance by up to 52.4 % under various operating conditions.
Tang et al [18]	Diverging/converging manifold microchannels	Water	Copper	0.03–0.27 mm	0.6 mm	Pumping power: 0.01–1.2 W, Heat flux: 500 W/cm ²	Optimized design reduced thermal resistance by 19.18 % at given pumping power, improving heat transfer and temperature uniformity.
Zhou et al [19]	Manifold microchannels with optimized pin fins	Water	Chrome-Zirconium Copper	0.5 mm	2 mm	Heat flux: 200–1000 W/cm ² , Flow rate: 0.5 m/s	Topological optimization enhanced heat transfer by up to 87 %, with a 60 % reduction in pressure drop.
Yan et al [20]	Y-shaped fractal microchannels	Water	Copper-based MCHS	0.5–1.5 mm	0.5 mm	Heat flux: 50–100 W/cm ² , Flow rate: 200–400 mL/min	Multi-objective optimization reduced thermal resistance by up to 40 % and improved energy efficiency by 25 %.
Sui et al [21]	Wavy microchannels	Water	Silicon	100–150 μ m	300 μ m	Heat flux: 1.5×10^6 W/m ² , Re: 100–800	Wavy channels significantly enhanced heat transfer by inducing chaotic advection, with a 153 % increase in heat transfer coefficient at Re = 800.
Zhang et al [27]	Rectangular microchannels with longitudinal vortex generators	Deionized Water	Silicon	300–450 μ m	100 μ m	Heat flux: 50–150 W/cm ² , Re: 100–1000	Optimization using the Taguchi method increased Nusselt number by 23.6 % and overall efficiency by 7.2 % compared to the original design.
Khodabandeh et al[28]	Double-layer microchannels with sinusoidal cavities and rectangular ribs	Water nanofluid/GNP-SDBS	Silicon	150 μ m	200 μ m	Heat flux: 50 W/cm ² , Re: 50–1000	The design improved Nusselt number by 20 % and reduced thermal resistance, enhancing heat transfer and temperature uniformity.
Yin et al [29]	Porous open microchannels	Deionized Water	Copper	0.4 mm	0.8 mm	Heat flux: 29.6–1871.1 kW/m ² , Mass flux: 151.5–551.1 kg/m ² s	Enhanced heat transfer coefficients in POM was up to 2× higher than SOM at high heat fluxes, with no local dryout observed even at highest heat fluxes.
Sun et al [30]	Microchannel heat sinks with Tesla valve, sector bump, and diamond bump	Water	Copper	0.6 mm	1.5 mm	Heat flux: 100 W/cm ² , Re: 400–1300	Novel designs increased <i>Nu</i> by up to 111.2 %, with sector bump showing the best overall thermal-hydraulic performance (<i>PEC</i> : 1.33–1.61)
Rui et al [31]	Microchannels with Tesla valve and sector bump	HFE7100 (Electronic Fluorinated Liquid)	Copper	1 mm	1 mm	Heat flux: 7.01–15.62 W/cm ² , Mass flow rate: 2.93–5.14 g/s	MCSB improved heat transfer by 45 % and reduced pressure drop by 25 % compared to MCTV.
Li et al [32]	3D Tesla valve manifold	Deionized Water	Copper	0.5 mm	0.2 mm	Heat flux: 375 W/cm ² , Flow rate: 45 mL/min	Tesla valve design improved temperature uniformity by 70 % and increased heat transfer coefficient by 119 % compared to straight channels.

distinctive design of the SYMTVM, derived from the symmetrical configuration of the TVM, needs a different inlet width for the fluid domains. Specifically, the inlet width ($W_{ch} = 0.62$ mm) of the SYMTVM is twice of the TVM ($W_{ch} = 0.31$ mm). The main parameters of the SYMTVM and TVM structures are shown in Table 2.

The single-stage valve structures of both TVM and SYMTVM, consisting of trunk and helix regions, are connected at an angle θ , as shown in Fig. 2.

These structures direct fluid through different ports, resulting in varied flow patterns. Forward flow is characterized by right-in and left-out trajectories, while reverse flow follows a left-in and right-out pattern, leading to a unidirectional valve effect. In forward flow, fluid travels predominantly through the trunk region, facilitating streamlined

flow. In contrast, reverse flow uses the inertia of the fluid to redirect it through the helix region, causing collisions with the trunk region fluid and creating obstructions.

The inherent design resistance of the Tesla valve is utilized in microchannel heat sinks, thereby boosting heat transfer and improving thermal efficiency. The increased backflow resistance results in a significant pressure drop, manifesting a 'fluid diode' effect. The effectiveness or duality (*Di*) of the Tesla valve is assessed by comparing pressure drops in reverse and forward flow, with *Di* typically ranging from 1 to 2 for micro/mini-channel structures, indicating significant differences in thermo-hydraulic performance between the two flow directions.

To comprehensively assess the thermo-hydraulic behavior of TVM and SYMTVM heat sinks, four comparative schemes are carried out in

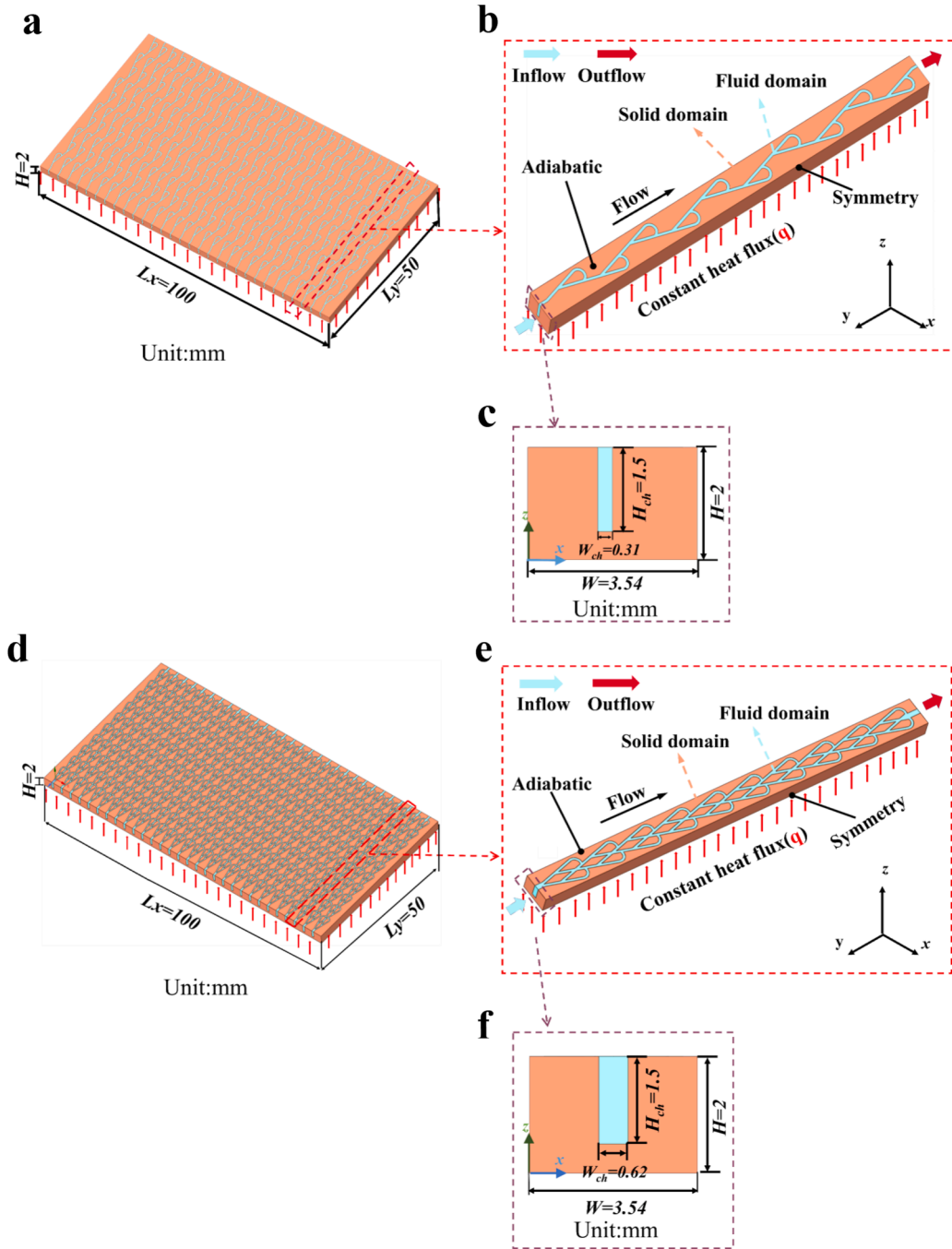


Fig. 1. (a) Schematic diagrams of the TVM. (b) The computational domain relevant to the TVM. (c) Cross-sectional view of TVM. (d) Schematic diagrams for the SYMTVM. (e) The computational domain of SYMTVM. (f) Cross-sectional view of SYMTVM.

Table 2
Structural parameters of TVM and SYMTVM.

Parameter	SYMTVM (unit:mm)	TVM (unit:mm)
L_x	100	100
L_y	50	50
H	2	2
W	3.54	3.54
W_{ch}	0.62	0.31
H_{ch}	1.5	1.5

this study, where the evaluation of the TVM and SYMTVM systems under the same mass flow conditions in both forward and reverse flow is undertaken, as illustrated in Fig. 3. Detailed geometric parameters for the TVM and SYMTVM heat sinks are presented in Fig. 4.

As a reference, this study also constructed a traditional rectangular microchannel heat sink (RM), as shown in Fig. 5. The width W of a computational unit for both the TVM and SYMTVM is 3.54 mm, in which six rectangular microchannels are contained, each with a width W_{ch} of 0.3 mm, a height H_{ch} of 1.5 mm, and a spacing W_b of 0.3 mm between them. The rectangular microchannel heat sink structure also exhibits periodicity. Therefore, to balance computational efficiency and accuracy, a single channel from the six available channels is selected as the computational unit for the rectangular microchannel heat sink. To

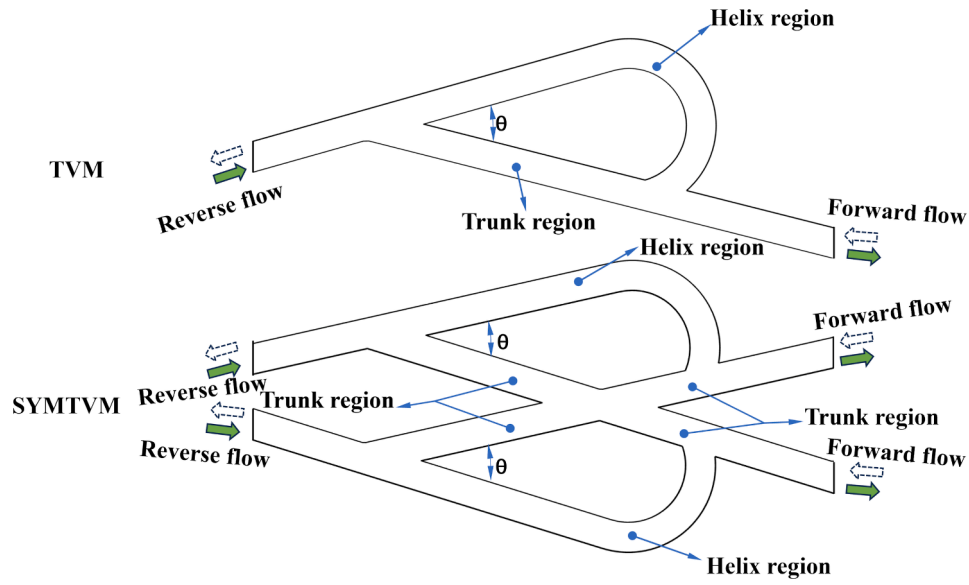


Fig. 2. Configuration of TVM and SYMTVM.

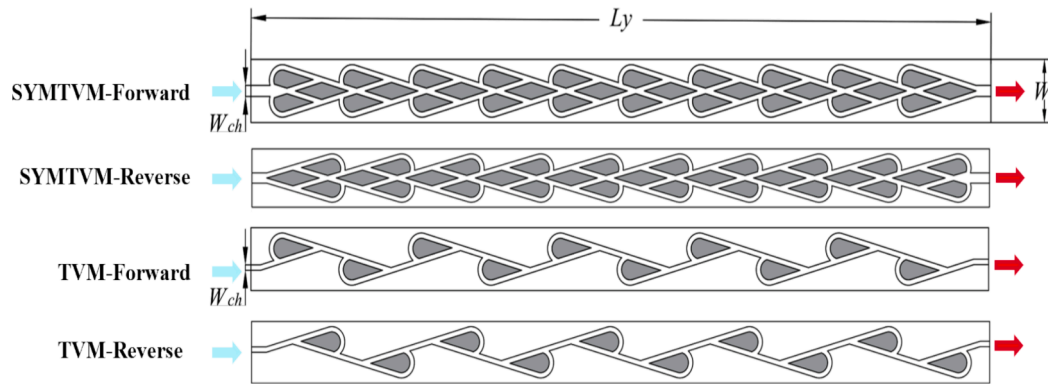


Fig. 3. Structure of SYMTVM-Forward, SYMTVM-Reverse, TVM-Forward and TVM-Reverse.

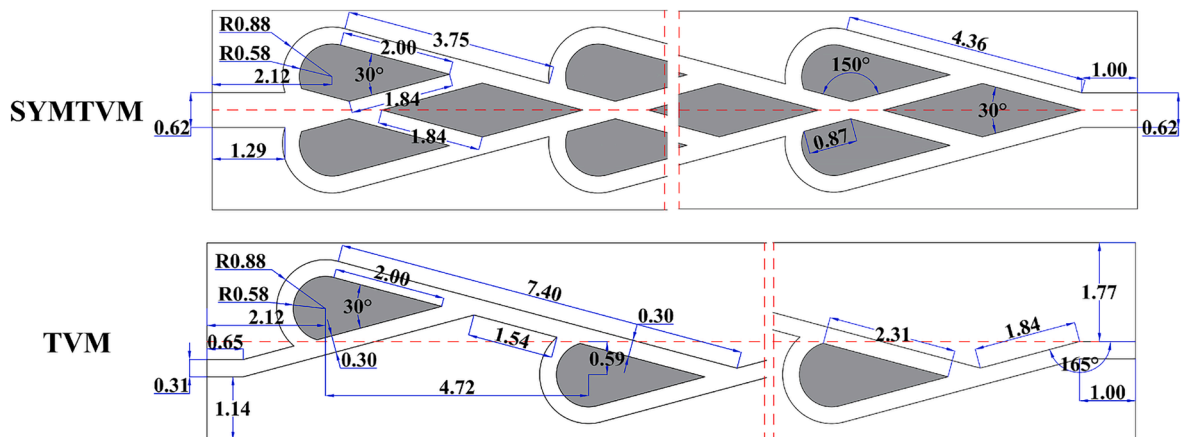


Fig. 4. Detailed parameters of SYMTVM and TVM (unit: mm).

ensure an equivalent comparison with the TVM and SYMTVM, the total inlet mass flow rate for the six rectangular channels is set to be the same as that for the TVM and SYMTVM, resulting in the inlet mass flow rate for a single rectangular microchannel being one-sixth of that for the TVM and SYMTVM.

2.2. Numerical method and boundary condition

Quantitative analyses of both forward and reverse fluid flow and heat transfer in Tesla Valve Microchannels (TVM) and Symmetric Tesla Valve Microchannels (SYMTVM) were performed, covering mass flow rates from 0.5459 to 1.6377 g/s. Correspondingly, the mass flow rate for

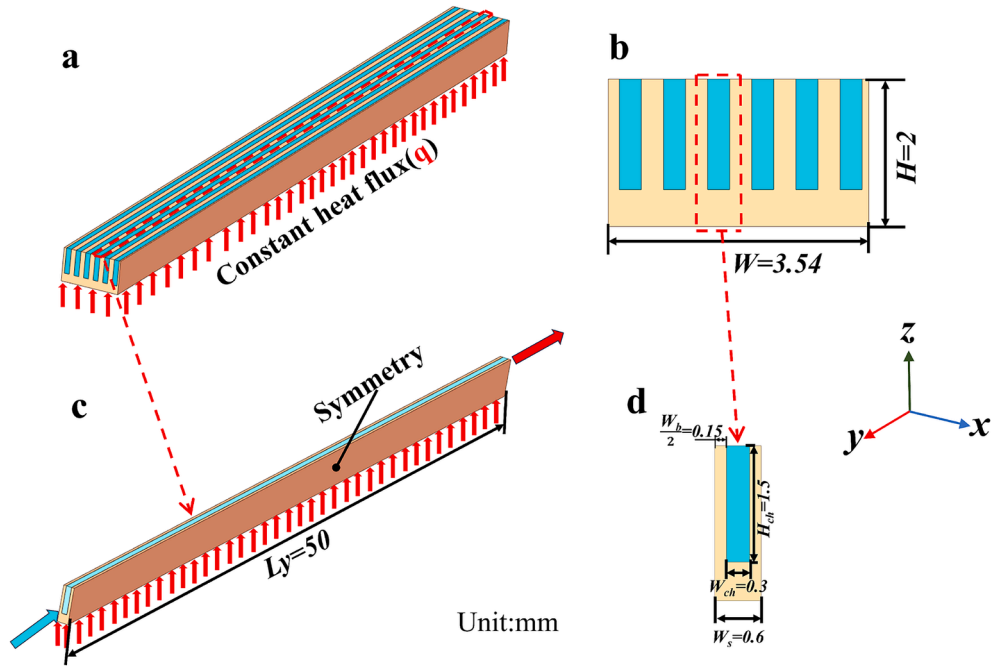


Fig. 5. (a) Schematic diagram of RM under the condition of a total width of 3.54 mm. (b) Cross-sectional view of RM under the condition of a total width of 3.54 mm. (c) Computational domain of the RM. (d) Cross-sectional view of the RM computational domain.

a single rectangular microchannel computational unit ranged from 0.09098 to 0.27295 g/s, with a maximum inlet Reynolds number of 300. Given that the fluid remained in the laminar flow regime under these conditions, a laminar flow model was employed to evaluate the thermodynamic performance of the traditional rectangular microchannel (RM). It is important to note that turbulence in Tesla Valve configured microchannels typically starts at Reynolds Numbers (Re) above 200 [41], which is attributed to the complex helical structures that induce localized flow separation and vortex near the channel walls. Traditional laminar flow models are unable to accurately represent the complex fluid dynamics and the associated thermal phenomena in turbulent or geometrically complex environments. In this investigation, the initial Reynolds number for TVM and SYMTVM at a baseline mass flow rate of 0.5459 g/s is calculated to be 585 and 513, respectively, indicating the onset of turbulence.

Given these complexities, the choice of the Realizable $k-\epsilon$ turbulence model with Enhanced Wall Treatment (EWF) was driven by its demonstrated ability to accurately capture the intricate flow phenomena [42], especially in cases of adverse pressure gradients and flow separation in this study. As illustrated in Fig. 6, various turbulence models and wall treatment approaches were compared, including the Standard Wall Function (StWF) and Scalable Wall Function (ScWF). The used experimental data in this study, from the results reported by Gamboa et al [43], served as a benchmark for validating our numerical simulations. The results show that the Realizable $k-\epsilon$ model with EWF provides the best consistence with the experimental data across a broad range of Reynolds numbers, particularly at higher values where flow dynamics become more intricate. The numerical models in this work were run under the conditions of the experiments to validate the accuracy. The Realizable $k-\epsilon$ model with EWF showed great superiority in predicting the Di values which is consistent with experimental results, outperforming other models in capturing the essential flow features. This highlights the model's effectiveness in balancing computational efficiency and predictive accuracy, making it the most suitable choice for the present work.

The numerical simulations were undertaken under the following assumptions:

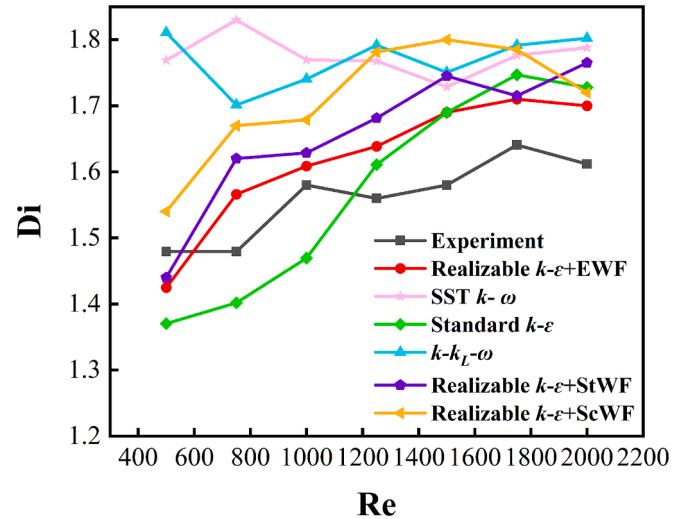


Fig. 6. Model validation by comparing with experimental results.

- (1) The fluid behaves as a Newtonian and incompressible medium;
- (2) The thermal and physical characteristics of the fluid are subject to change in response to alterations in temperature;
- (3) The processes of flow and heat transfer are assumed to maintain a steady-state condition;
- (4) Radiative heat transfer and gravitational effects are both deemed insignificant.

Taking the assumptions into consideration, the governing equations were formulated using the realizable $k-\epsilon$ turbulence model to address turbulent flow,

Continuity equation:

$$\nabla \cdot (\rho_f \bar{V}) = 0 \quad (1)$$

Momentum equation:

$$\bar{V} \cdot \nabla (\rho_f \bar{V}) = -\nabla P + \left(\mu_f \nabla \bar{V} \right) \quad (2)$$

Energy equation:

$$\bar{V} \cdot \nabla (\rho_f c_p T_f) = \nabla \cdot (k_f T_f) \quad (3)$$

Where $\bar{V}$ represents the velocity vector, P denotes the pressure, T_f stands for the fluid temperature, c_p signifies the specific heat capacity of the fluid, and k_f indicates the fluid's thermal conductivity.

The energy equation in the solid domain is expressed as,

$$\nabla \cdot (K_s \nabla T_s) = 0 \quad (4)$$

Where K_s denotes the thermal conductivity of the solid material, while T_s represents the temperature.

The transport equations for turbulent kinetic energy (k) and its dissipation rate (ε) within the framework of the realizable k - ε model are formulated as follows[44],

$$\frac{\partial \varepsilon}{\partial t} (\rho k) + \frac{\partial}{\partial x_j} (\rho k u_j) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \varepsilon - Y_M + S_k \quad (5)$$

$$\begin{aligned} \frac{\partial \varepsilon}{\partial t} (\rho \varepsilon) + \frac{\partial}{\partial x_j} (\rho \varepsilon u_j) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] &+ \rho C_1 S \varepsilon - \rho C_2 \frac{\varepsilon^2}{k + \sqrt{\nu \varepsilon}} + C_{1\varepsilon} \frac{\varepsilon}{k} C_{3\varepsilon} G_b + S_\varepsilon \end{aligned} \quad (6)$$

Where $C_{1\varepsilon} = 1.44$, $C_2 = 1.9$, $\sigma_k = 1.0$, $\sigma_\varepsilon = 12$. C_1 can be calculated by the following equation,

$$C_1 = \max \left[0.43, \frac{\eta}{\eta + 5} \right], \eta = S \frac{k}{\varepsilon}, S = \sqrt{2 S_{ij} S_{ij}} \quad (7)$$

Boundary conditions for the computational domain are outlined as follows: (1) The inlet of the microchannel has a uniform mass flow and the fluid temperature is constant at 300 K; (2) The outlet pressure of the microchannel is set to be 0 Pa; (3) The solid domain's bottom wall is subjected to a constant heat flux of 10^6 W/m²; (4) No slippage and continuous temperature is assumed at the solid-liquid interface; (5) Symmetry boundary conditions govern both lateral faces of the solid domain; (6) Other surfaces of the solid are all treated with adiabatic boundary conditions; (7) For the top surface of the fluid domain, both adiabatic and no-slip boundary conditions are implemented.

In this work, copper served as the substrate for the microchannel, and the material properties are listed in Table 3. Deionized water, chosen for its temperature-dependent properties as highlighted in Table 4[45], serves as the working fluid within the microchannel to enhance accuracy.

Fluid dynamics and thermal properties were evaluated across microchannel models using Fluent (2022R1), with the mass flow rates from 0.5459 g/s to 1.6377 g/s. The coupling of pressure and velocity was facilitated by the standard SIMPLE algorithm. The pressure equation was resolved using the standard format, while the momentum and energy equations were discretized utilizing the second-order windward technique. Verification of the solution's convergence occurred when the normalized residuals decreased to be below 10^{-5} for the continuity equation, dipped under 10^{-6} for the momentum equation, and fell below 10^{-9} for the energy equation.

Table 3
Physical parameters of copper.

Material	Density (kg/m ³)	Specific heat (J/(kg·K))	Thermal conductivity(W/(m·K))
Copper	8978	381	387.6

2.3. Mesh sensitivity

To balance accuracy with computational efficiency, a comprehensive grid sensitivity analysis was performed on the TVM and SYMTVM heat sink models, aiming for grid independence at a mass flow rate of 1.0918 g/s. Grid density ranged from 600,000 to 3,260,000 cells for the TVM heat sink and from 580,000 to 4,800,000 cells for the SYMTVM heat sink, as detailed in Table 5 and Table 6. The escalation in grid size from 1,846,186 to 3,261,266 resulted in variations in the fluid pressure drop and the maximum temperature at the lower surface of the solid domain within the TVM heat sink that were less than 0.75 %. For the SYMTVM heat sink, variations in fluid pressure drop and maximum temperature at the lower surface of the solid domain were minimal, less than 0.97 %, as the grid numbers increased from 2,816,560 to 4,828,512. To make full use of computing resources without compromising accuracy, the simulations for both models were conducted using the respective fourth grid sets: 1,846,186 cells for the TVM and 2,816,560 cells for the SYMTVM. In the grid independence examination for the rectangular microchannel heat sink, the grid density ranged from 200,000 to over 3,000,000 cells, as shown in Table 7. As the grid size increased beyond 1,884,186 cells, variations in pressure drop and maximum temperature were minimal, with differences of less than 0.38 % and 0.07 %, respectively. Therefore, the fifth grid set with 2,454,670 cells was selected to ensure accurate results while optimizing computational resources. Fig. 7 illustrates the meshing results of the TVM, SYMTVM, and RM under the selected scheme, along with a local zoomed view.

2.4. Data reduction

To compare thermal performance, the microchannel heat sinks were tested under identical total mass flow rates. The Reynolds number (Re) can be calculated using the following equation,

$$Re = \frac{\rho_f u_{in} D_h}{\mu_f} \quad (8)$$

Where ρ_f represents the density of deionized water, and μ_f denotes its dynamic viscosity. The term u_{in} refers to the average velocity at the inlet, while D_h indicates the microchannel's hydraulic diameter, which is ascertainable through the following equation,

$$D_h = \frac{2(W_{ch} H_{ch})}{(W_{ch} + H_{ch})} \quad (9)$$

The overall pumping power (Φ_{tot}) required for the microchannel heat sink is represented by Equ.(10).

$$\Phi_{tot} = u_{ave} A_{in} \Delta P \quad (10)$$

Where u_{ave} denotes the average velocity and A_{in} represents the total area of the inlet, with ΔP indicating the average pressure drop. When the mass flow rate is identical in both microchannel heat sinks, the pressure drops provide an estimate of the pumping power.

The temperature difference on the heat source surface δT is calculated as the difference between its maximum and minimum temperatures, as detailed in Equ.(11).

$$\delta T = T_{max} - T_{min} \quad (11)$$

Where T_{max} and T_{min} denote the maximum and minimum temperatures on the surface of the heat source, respectively.

The formula for calculating the effective thermal resistance R_t is given by,

$$R_t = \frac{T_{max} - T_{min}}{q_w} \quad (12)$$

The average heat transfer coefficient can be computed as delineated below,

Table 4

The physical properties of water as a function of temperature.

Parameters	Equation (T in $^{\circ}\text{C}$ for ρ_f , in K for μ_f , $c_{p,f}$ and k_f)
Density	$\rho_f = \frac{(998.84 + 18.225T - 7.92 \times 10^{-3}T^2 - 5.545 \times 10^{-5}T^3 + 1.498 \times 10^{-7}T^4 - 3.933 \times 10^{-10}T^5)}{1 + 1.816 \times 10^{-2}T}$
Dynamic viscosity	$\mu_f = 2.414 \times 10^{-5} \times 10^{\frac{247.8}{T-140}}$
Specific heat	$c_{p,f} = 8958.9 - 40.535T + 0.11243T^2 - 1.014 \times 10^{-4}T^3$
Thermal conductivity	$k_f = -0.58166 + 6.3556 \times 10^{-3}T - 7.964 \times 10^{-6}T^2$

Table 5

Grid independence examination for TVM.

Case	No. of elements	Pressure drop (ΔP)	Maximum temperature of heat source surface temperature(K)
1	600,946	59000.66	355.73
2	1,088,591	56753.05	356.42
3	1,562,164	56507.20	356.49
4	1,846,186	56352.49	356.55
5	2,366,599	56134.19	356.62
6	3,261,266	55931.79	356.68

Table 6

Grid independence examination for SYMTVM.

Case	No. of elements	Pressure drop (ΔP)	Maximum temperature of heat source surface temperature(K)
1	582,696	14831.806	355.18906
2	1,291,960	14561.22	355.23123
3	1,855,977	14481.584	355.26398
4	2,816,559	14458.064	355.26663
5	3,632,458	14443.015	355.27
6	4,828,512	14440.217	355.272

Table 7

Grid independence examination for RM.

Case	No. of elements	Pressure drop (ΔP)	Maximum temperature of heat source surface temperature(K)
1	201,487	4091.23	360.37625
2	724,763	3681.7	361.7754
3	1,368,420	3312.59	362.715
4	1,884,186	3270.68	362.812
5	2,454,670	3243.6309	362.86923
6	3,076,523	3231.25	362.8701

$$h_{av} = \frac{qL_y W}{A_{sf}(T_{w,ave} - 0.5(T_{in} - T_{out}))} \quad (13)$$

Where A_{sf} denotes the fluid–solid interface area, $T_{w,ave}$ indicates the bottom wall's average temperature in the solid domain, T_{in} and T_{out} are the average temperatures at the microchannel inlet and outlet, respectively.

The Nusselt number (Nu) functions as a measure of the heat transfer efficiency of the microchannel and can be determined by the following formula,

$$Nu = \frac{h_{av} D_h}{k_{f,ave}} \quad (14)$$

Where $k_{f,ave}$ denotes the thermal conductivity of the working fluid.

To assess both the overall and localized frictional resistance of SYMTVM and TVM, the average Darcy friction factor (f) is determined using the following formula,

$$f = \frac{2\Delta P D_h}{\rho_f u_{in}^2 L} \quad (15)$$

The comprehensive evaluation of the heat sinks' performance is

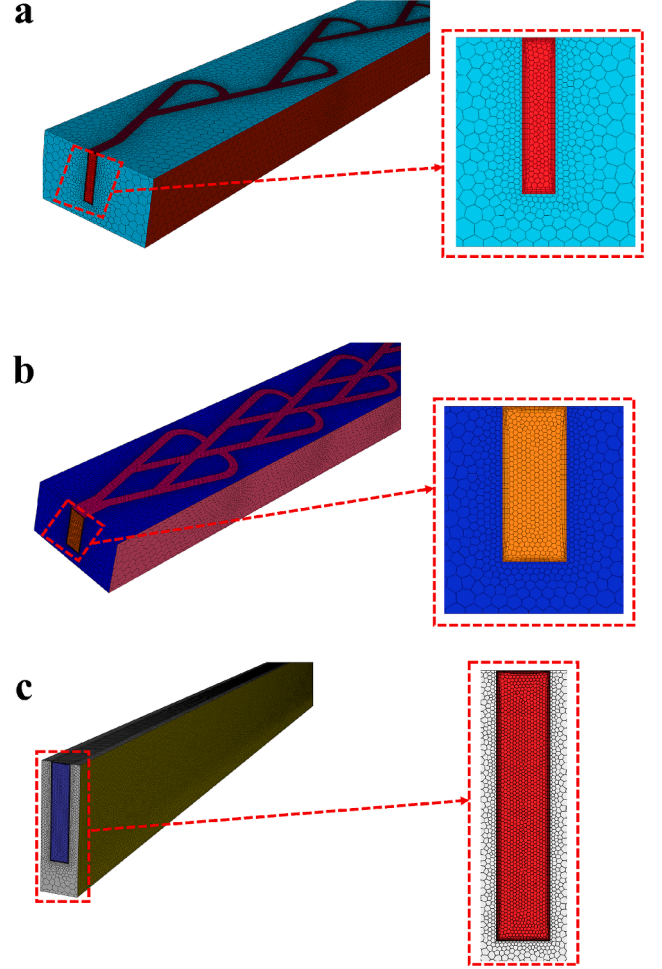


Fig. 7. (a) Total delineation results of TVM under 1,846,186 meshes and local zoomed view. (b) Total delineation results of SYMTVM under 2,816,560 meshes and local zoomed view. (c) Total delineation results of RM under 2,454,670 meshes and local zoomed view.

conducted through a particular performance criterion, the Performance Evaluation Criteria (PEC), which is expressed in Equ.(16).

$$PEC = \frac{(Nu/Nu_0)}{(\Delta P/\Delta P_0)^{1/3}} \quad (16)$$

Pressure diodicity (Di) is the pressure drop ratio in reverse and forward directions at a specific velocity, and it is represented by the ratio in Eq. (17).

$$Di = \frac{\Delta P_r}{\Delta P_f} \quad (17)$$

Temperature inhomogeneity (α) is calculated by dividing the maximum temperature difference across the wall by the average wall temperature, which can be given by,

$$\alpha = \frac{T_{\max} - T_{\min}}{T_{\text{ave}}} \quad (18)$$

2.5. Validation of numerical results

To confirm the accuracy of the numerical method, a computational model was built following the experimental framework outlined by Gamboa et al [43], with the same boundary conditions applied. Gamboa et al conducted experiments to optimize Tesla valve shapes, focusing on enhancing diodicity by measuring pressure drops under steady-flow conditions across various valve designs where the valves at Reynolds numbers up to 2000 were tested and a 25 % improvement in diodicity was achieved compared to conventional designs. The results were applied to validate the computational models in this study, to confirm the accuracy of our numerical simulations for microchannel heat sinks. As shown in Fig. 8, the Di values along with the pressure drop throughout the Tesla valve channel for Reynolds numbers ranging from 500 to 2000 were examined. The simulation results were of good consistency with the experimental data, with a maximum deviation of 13.1 % and an average discrepancy of 7.03 % at $Re = 1500$. As the Reynolds number decreased to 1000, the relative error for Di values and pressure drops narrows to be about 5 %, thus validating the precision and dependability of the proposed numerical model.

To show the reliability of the rectangular microchannel heat sink simulations, a similar model from the literature was selected for comparative analysis. Specifically, the fluid flow and heat transfer in the rectangular microchannel were simulated and compared against experimental data reported by Ho et al [46]. The results, as shown in Fig. 9, indicate that the difference remains within a reasonable range while there is a certain degree of discrepancy between the experimental data and simulation results. The potential sources of error include the selection of the mathematical model, differences in mesh quality, and so on. Overall, these findings confirm the validity of the simulation approach.

3. Results and discussion

3.1. Heat transfer characteristics of TVM and SYMTVM

The heat transfer capabilities of SYMTVM and TVM heat sinks under varying flow directions in single-phase flow were firstly examined. Fig. 10 presents a comprehensive analysis of the key thermal parameters measured at the heated surface under varying mass flow rates, including the maximum temperature ($T_{\max}$), minimum temperature ($T_{\min}$), temperature difference (δT), and temperature inhomogeneity (α). A comparison of the aforementioned parameters reveals a clear inverse relationship between mass flow rate and both $T_{\max}$ and $T_{\min}$. This

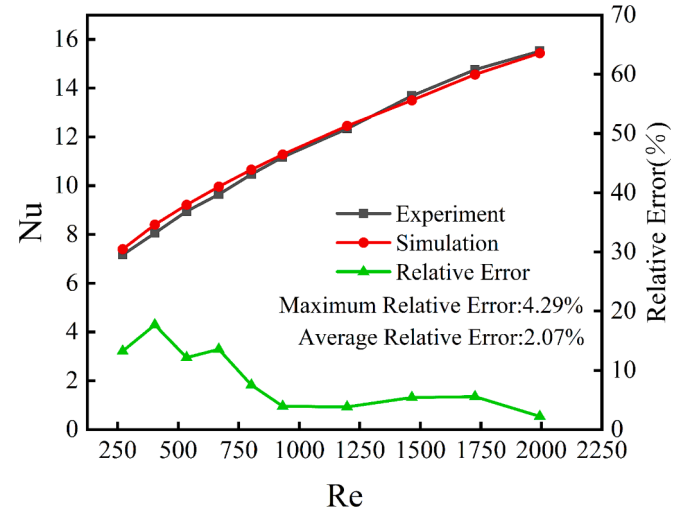


Fig. 9. Comparison of the numerical results with the experimental data from Ho et al [46].

indicates that higher mass flow rates contribute to enhanced heat dissipation efficiency. The improvement in heat dissipation is primarily due to an increased convective heat transfer coefficient, which remains effective even when the heat flux is unchanged. For constant inlet temperature and heat flux, $T_{\max}$ and effective thermal resistance (R_e) follow similar trends with increasing mass flow rate, as described in Eq.12. Fig. 10(a) shows that under identical inlet flow conditions, the lowest $T_{\max}$ is obtained in the SYMTVM under reverse flow, highlighting the SYMTVM's superior heat transfer efficiency in this special setup. When operating in reverse flow, both SYMTVM and TVM show a reduction in $T_{\max}$ compared to forward flow, indicating improved heat transfer coefficients in reverse flow conditions. In addition, the difference in $T_{\max}$ between forward and reverse flow for SYMTVM and TVM heat sinks reaches maximum at 8.75 K and 12.48 K at a mass flow rate of 0.5459 g/s, respectively. At higher mass flow rates, there is a notable reduction in the $T_{\max}$ difference between the forward and reverse flows in both SYMTVM and TVM heat sinks. The $T_{\max}$ difference for forward and reverse flow in SYMTVM and TVM heat sinks is 7.042 K and 7.93 K, respectively.

Fig. 10(b) shows that the characteristics of the minimum temperature ($T_{\min}$) are different from $T_{\max}$. The lowest $T_{\min}$ is obtained in SYMTVM-Forward, followed by SYMTVM-Reverse. $T_{\min}$ for TVM is higher than these two scenarios in both forward and reverse flow. Fig. 10 (c) shows that the temperature difference (δT) across the heating surface is a crucial metric for assessing the uniformity of chip cooling. Over all

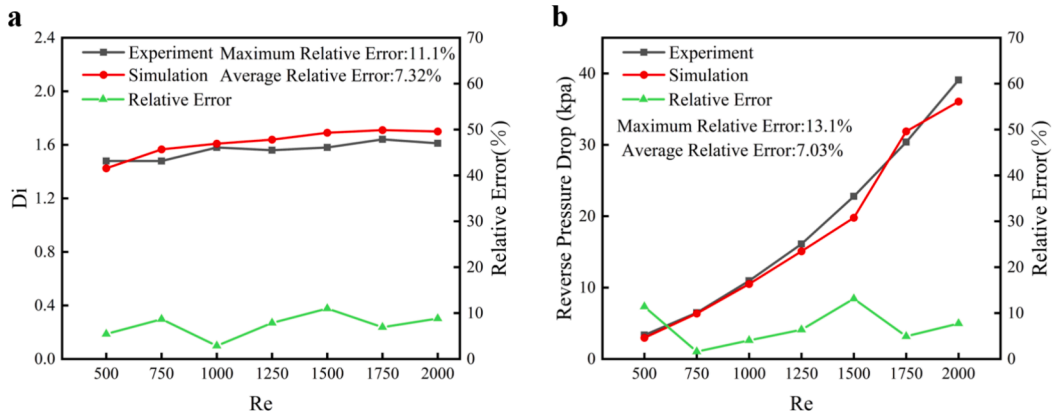


Fig. 8. Validation of the simulation model: (a) Comparative analysis of Di values for the Tesla valve from experimental and simulation data. (b) Comparative assessment of pressure drop between experimental and simulation results [43].

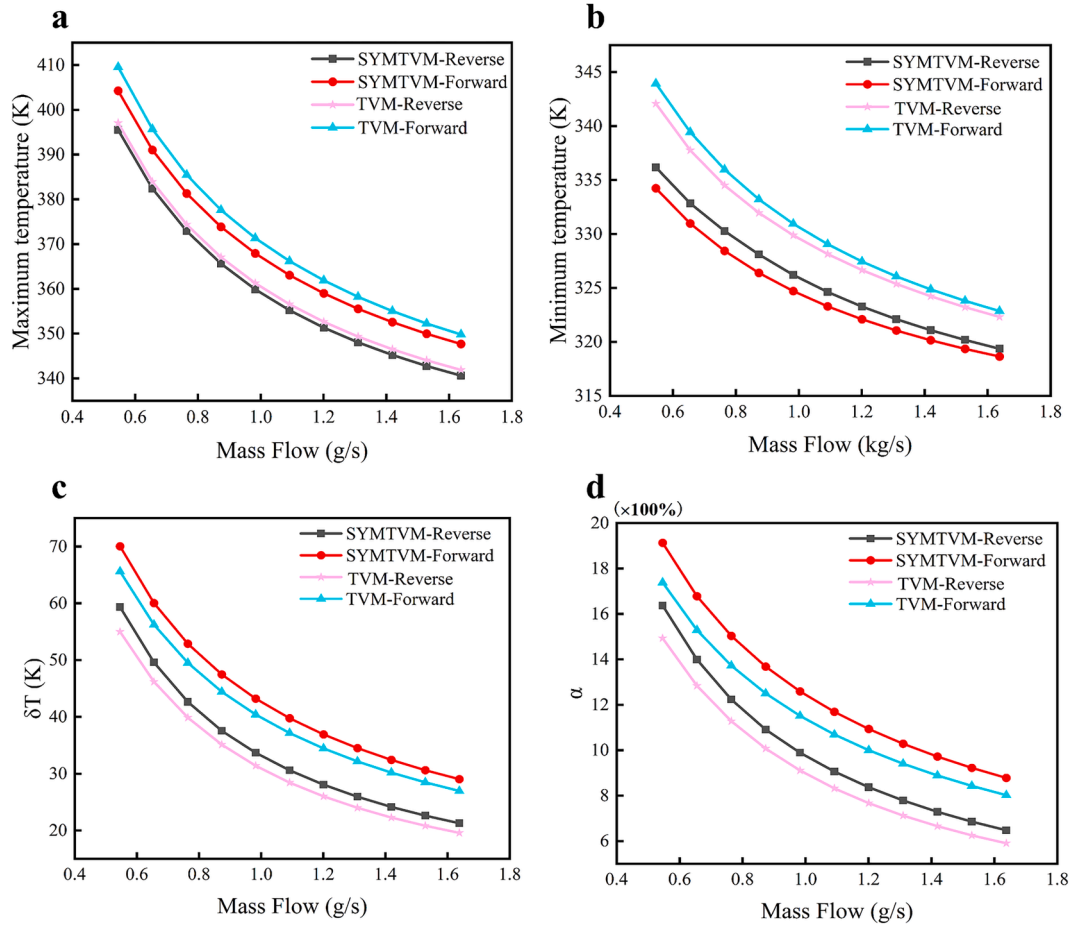


Fig. 10. (a) T_{max} . (b) T_{min} . (c) δT . (d) α variations under varying inlet mass flow rates.

mass flows, SYMTVM-Forward has the highest δT , ranging from 70.02 K to 29.0239 K. However, the temperature uniformity is significantly better in SYMTVM-Reverse, with δT values ranging from 59.32 K to 21.26 K. TVM-Reverse is of the lowest temperature difference. Fig. 10(d) illustrates that the parameter α decreases as the mass flow rate increases. Under reverse flow, both the SYMTVM and TVM exhibit reduced temperature inhomogeneity in comparison with forward flow. In the SYMTVM-reverse setup, the parameter α fluctuates between 6.4 % and

16.3 %, while it spans from 5.9 % to 14.9 % in the two-layer reverse flow arrangement. This illustrates that the reverse flow enhances temperature uniformity in both SYMTVM and TVM systems. Reduced temperature differences lead to decreased thermal stress on components, potentially improving the reliability of thermal performance.

Fig. 11 depicts the temperature fluctuations recorded on the heat source surface across four distinct scenarios, with the inlet mass flow rate fixed at 1.091 g/s. As the fluid progresses through the

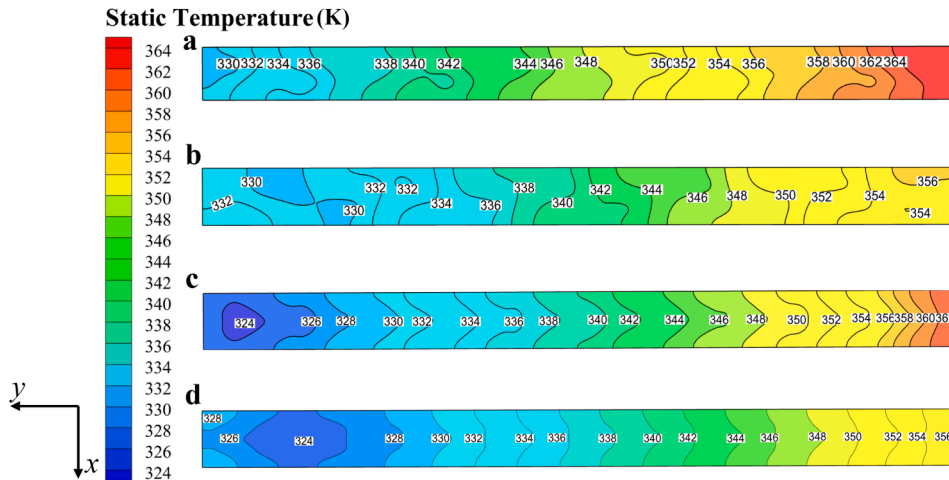


Fig. 11. The influence of fluid entry direction and structural design on the distribution of wall temperatures under an inlet mass flow rate of 1.091 g/s: (a) TVM-Forward. (b) TVM-Reverse. (c) SYMTVM-Forward. (d) SYMTVM-Reverse.

microchannels, its temperature gradually rises. Consequently, the highest temperatures are typically found near the microchannel outlet, whereas the lowest temperatures are near the microchannel inlet. However, the inlet consists of a straight channel in SYMTVM and TVM heat sinks, where the fluid flows in a laminar state, resulting in limited fluid mixing and suboptimal localized heat transfer. In the regions formed by flow separation and vortex generation along the wall of SYMTVM and TVM heat sinks, the mixing of near-wall thermal fluid with the mainstream fluid becomes more effective after entering the manifold structure. This enhancement is attributed to structural heat transfer processes, including conduction, convergence, and collision. As a result, the temperature in these areas is lower than the inlet, indicating improved local thermal exchange between the fluid and the solid surfaces.

Fig. 12 highlights the superior thermal performance of the SYMTVM-Reverse configuration, which achieves the lowest peak temperature and most uniform temperature distribution. This efficiency is attributed to enhanced fluid mixing and vortex generation at the interconnection angle which disrupt the thermal boundary layer and improve convective heat transfer. In comparison, the TVM configurations exhibit higher maximum temperatures and less uniform temperature profiles, especially in reverse flow, where flow separation increases thermal resistance without sufficiently enhancing heat transfer. The SYMTVM-Forward outperforms the TVM designs, but the SYMTVM-Reverse delivers even better results by maximizing cooling through increased flow disturbances.

3.2. Flow characteristics of TVM and SYMTVM

Fig. 13 depicts the comprehensive pressure drop (ΔP) from the inlet to the outlet across different scenarios. The highest pressure drop in TVM-Reverse occurs with an increase of the inlet mass flow rate. At the peak flow rate of 1.637 g/s, the pressure drop in TVM-Reverse exceeds that in TVM-Forward, SYMTVM-Forward, and SYMTVM-Reverse by factors of 2.48, 5.87, and 4.07, respectively. Furthermore, the pressure drop observed in the TVM, whether in the forward or reverse direction, is notably higher compared to that in the SYMTVM at equivalent mass flow rates. The variation is ascribed to the increased fluid velocity in TVM for an equivalent mass flow rate, leading to a more significant pressure drop.

In TVM-Forward and SYMTVM-Forward, the coolant mainly flows through the trunk channel, whereas it is mainly directed through the helix channel in TVM-Reverse and SYMTVM-Reverse. To analyze flow characteristics in these heat sinks, velocity cloud plots at the mid-plane

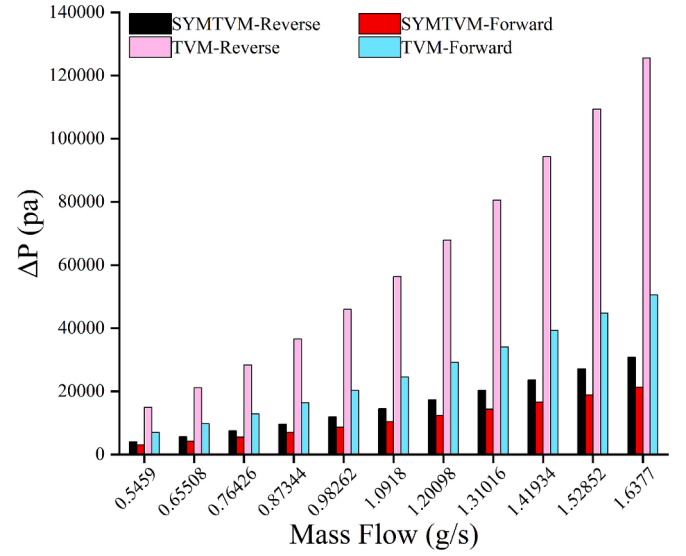


Fig. 13. Changes of the overall pressure drop (ΔP) with different inlet mass flow rates.

level within the Tesla valve cells of the second and third stages are shown in Fig. 14. In the TVM-Reverse and SYMTVM-Reverse flow fields, the inertia of the coolant predominantly leads to the separation into the helix region at the bifurcation point. The coolant flows then converge in the straight and annular passages, resulting in a jet impingement event characterized by high water flow velocities impinging on the channel walls. The complex interaction among mixing processes, self-impact phenomena, and secondary flows results in a significant enhancement of the heat transfer efficiency within the system. In contrast, the dynamics within the TVM-Forward and SYMTVM-Forward flow fields exhibit a relatively simpler nature. In this context, the cooling water traverses a route characterized by incremental changes in direction within the bifurcation area, leading a substantial portion of the diverged flow towards the linear channel. Moreover, the generation of vortices within the bifurcation area induces secondary flows that interact with the boundary layer, thereby enhancing heat transfer efficiency.

Fig. 15 and Fig. 16 demonstrates the emergence of localized transverse vortices at the junction of the trunk/helix outlets in TVM-Forward and SYMTVM-Forward, occurring as the flow divides during forward flow. Fig. 15a shows low speed vortex near the helix outlet, where the

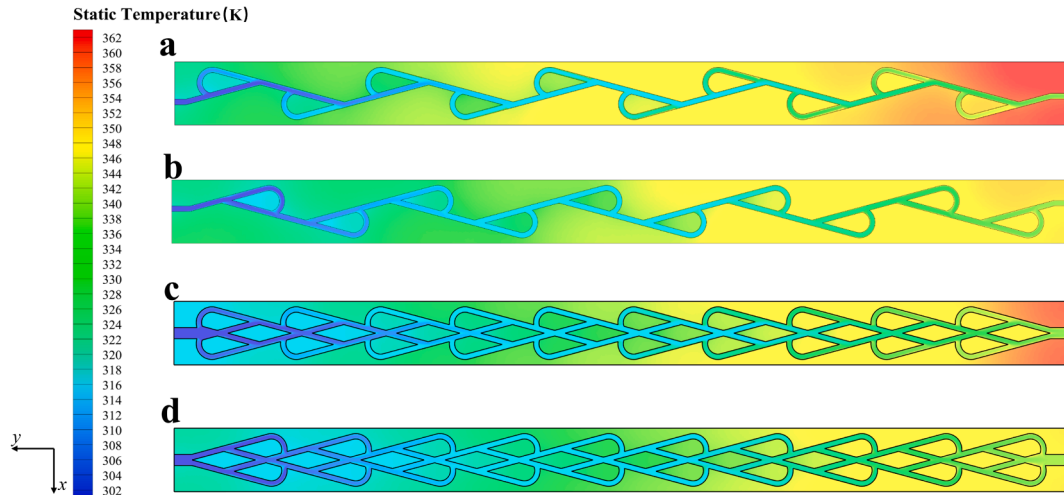


Fig. 12. The temperature contours at $z = 1$ mm and an inlet mass flow rate of 1.091 g/s: (a) TVM-Forward. (b) TVM-Reverse. (c) SYMTVM-Forward. (d) SYMTVM-Reverse.

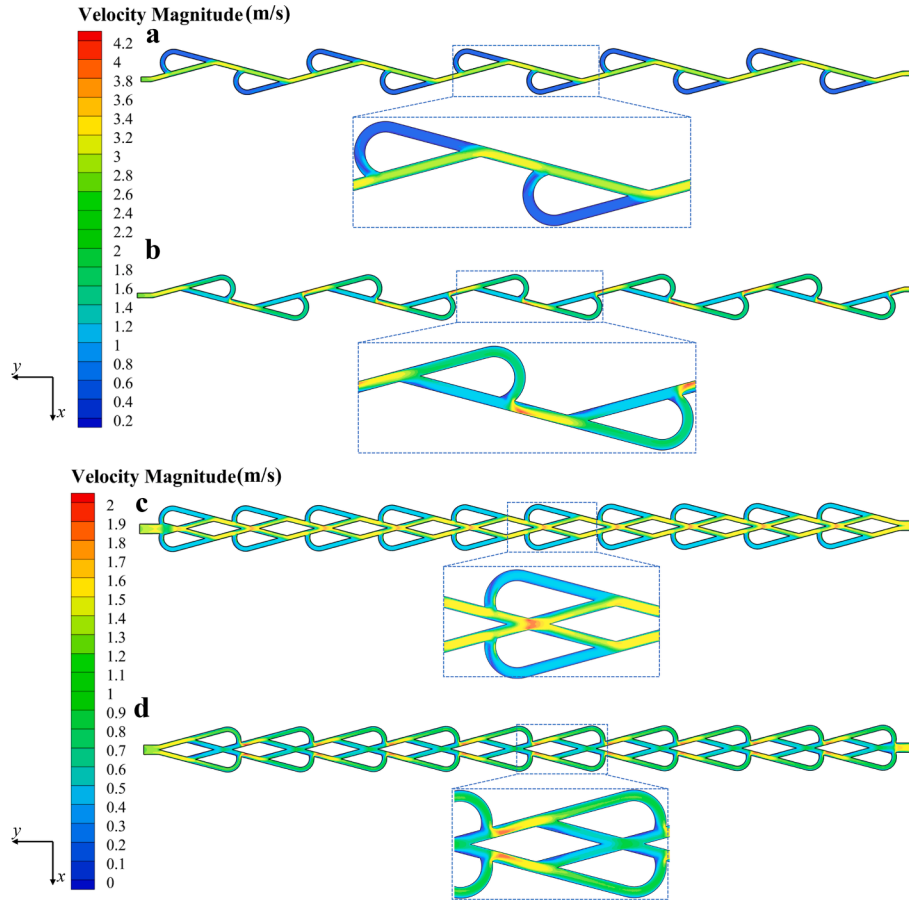


Fig. 14. Velocity contours at an inlet mass flow rate of 1.091 g/s: (a) TVM-Forward. (b) TVM-Reverse. (c) SYMTVM-Forward. (d) SYMTVM-Reverse.

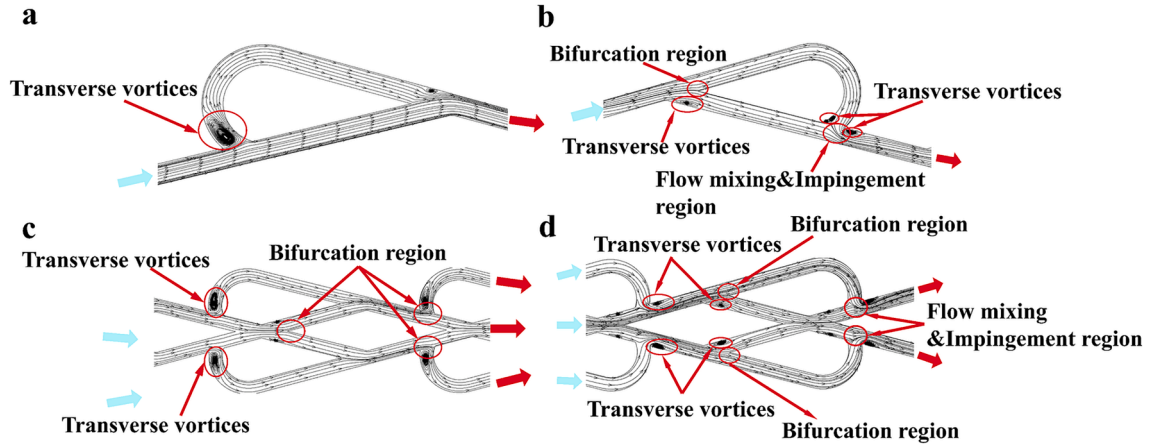


Fig. 15. Streamline distributions at $z = 1$ mm and an inlet mass flow rate of 1.091 g/s: (a) TVM-Forward. (b) TVM-Reverse. (c) SYMTVM-Forward. (d) SYMTVM-Reverse.

local flow becomes unstable and takes on a more multidirectional nature to improve heat transfer. These transverse vortices enhance convective heat transfer from the channel's center to its walls. More complex flow phenomena appear in TVM-reverse and SYMTVM-reverse, as shown in Fig. 15b. Particularly, small vortices form after each instance of flow separation, and additional vortices develop due to self-impingement and the flow mixing near the spiral outlet. The creation of these vortices induces unsteady, multidirectional fluid flow, leading to significant fluid mixing and improved thermal efficiency. Notably, the SYMTVM contains more vortices and bifurcations within its single-stage structure

than the TVM. As the fluid navigates through these bifurcation structures, it undergoes repeated divergence and convergence. This process leads to periodic disturbances and continuous regeneration of the thermal boundary layer, thereby leading to a notable enhancement in heat transfer efficiency.

In both TVM and SYMTVM, fluid divergence occurs in each Tesla valve unit, especially in the divergence and convergence areas. This results in enhanced heat transfer where flow fluctuations develop and intensify upon contact with the walls to form vortices, as shown in Fig. 15. Turbulent Kinetic Energy (*TKE*) is defined, which indicates the

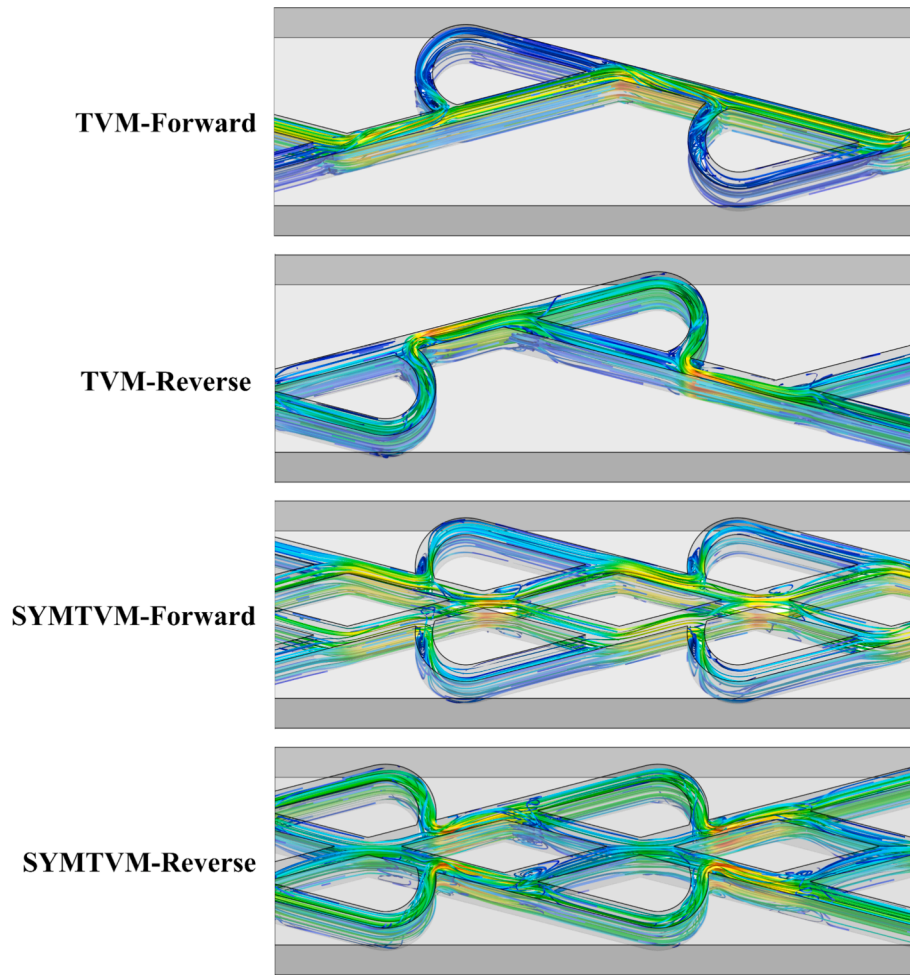


Fig. 16. 3D Streamline distributions at $z = 1$ mm and an inlet mass flow rate of 1.091 g/s.

intensity of flow pulsations, as illustrated in Fig. 17. The highest turbulence kinetic energy (*TKE*) and flow fluctuations are found primarily in the areas where the Tesla valve unit widens and narrows. Within these diverging and converging regions, strong shocks, flow mixing and localized acceleration are prevalent. These phenomena contribute to increased local velocities, enhanced flow interactions, and boundary layer disturbances, which collectively increase the heat transfer rates in the Tesla valve microchannels. TVMs, when operating at elevated fluid velocities, exhibit significantly increased levels of turbulent kinetic energy (*TKE*), resulting in better heat transfer efficiency. Specifically, the reverse flow in both TVM and SYMTVM shows a marked increase in turbulent kinetic energy (*TKE*) intensity and more significant flow fluctuations than those seen in forward flow. In scenarios involving reverse flow, the convergence zone experiences a significant increase in turbulent kinetic energy, accompanied by more pronounced turbulent disturbances. This effect leads to a marked improvement in the cooling effectiveness of the system. In terms of flow resistance, it has been noted that within the Tesla valve channel, the fluid tends to bypass the bifurcation points in the curved structure during forward flow. Only a small portion of the coolant flows into the convergence at angles less than 90° , ensuring minimal resistance. This means that, in the case of forward flow, both TVM and SYMTVM can direct more coolant in the trunk channel thus reducing flow resistance. In reverse flow conditions, coolant moves through curved channels with convergence angles over 90° , leading to backflow collisions which improve cooling performance but lead to increased flow resistance and pressure drop within the channel.

It should be noted that the optimization of TVM and SYMTVM

channels focuses on delivering excellent heat transfer performance while keeping pressure drop to a minimum within the channels. Achieving this balance is essential for the overall improvement and operational efficiency of these microchannel cold plates.

3.3. Comprehensive performance of TVM and SYMTVM

Fig. 18 depicts how the average heat transfer coefficient and the combined performance factor vary across different flow regimes. From Fig. 18(a), it is clear that a rise in the inlet mass flow rate results in an elevated average heat transfer coefficient. This demonstrates a direct positive correlation between the inlet mass flow rate and the efficiency of heat transfer. Specifically, in SYMTVM, the heat transfer coefficients for reverse flow range from 1.106 to 1.117 times those for forward flow. For TVM, the reverse flow coefficients are approximately 1.267 to 1.285 times those for forward flow. A higher coefficient of TVM than SYMTVM at the same mass flow rates is caused by the smaller inlet area of TVM, which results in higher fluid velocity.

Enhancing the design of microchannel heatsinks is directed towards enhancing thermal efficiency while reducing pressure drop. This optimization is subjected to critical evaluation utilizing the Performance Evaluation Criteria (*PEC*). The proper functioning of the *PEC* is crucial for ensuring effective and efficient thermal management. High *PEC* values indicate excellent performance. As illustrated in Fig. 18(b), SYMTVM-Reverse consistently demonstrates superior performance in terms of *PEC* values compared to TVM-Reverse across a range of inlet mass flow rates. In particular, it is found that SYMTVM-Reverse achieves *PEC* values that are 95.7 %-96.2 % higher than those of TVM-Reverse in

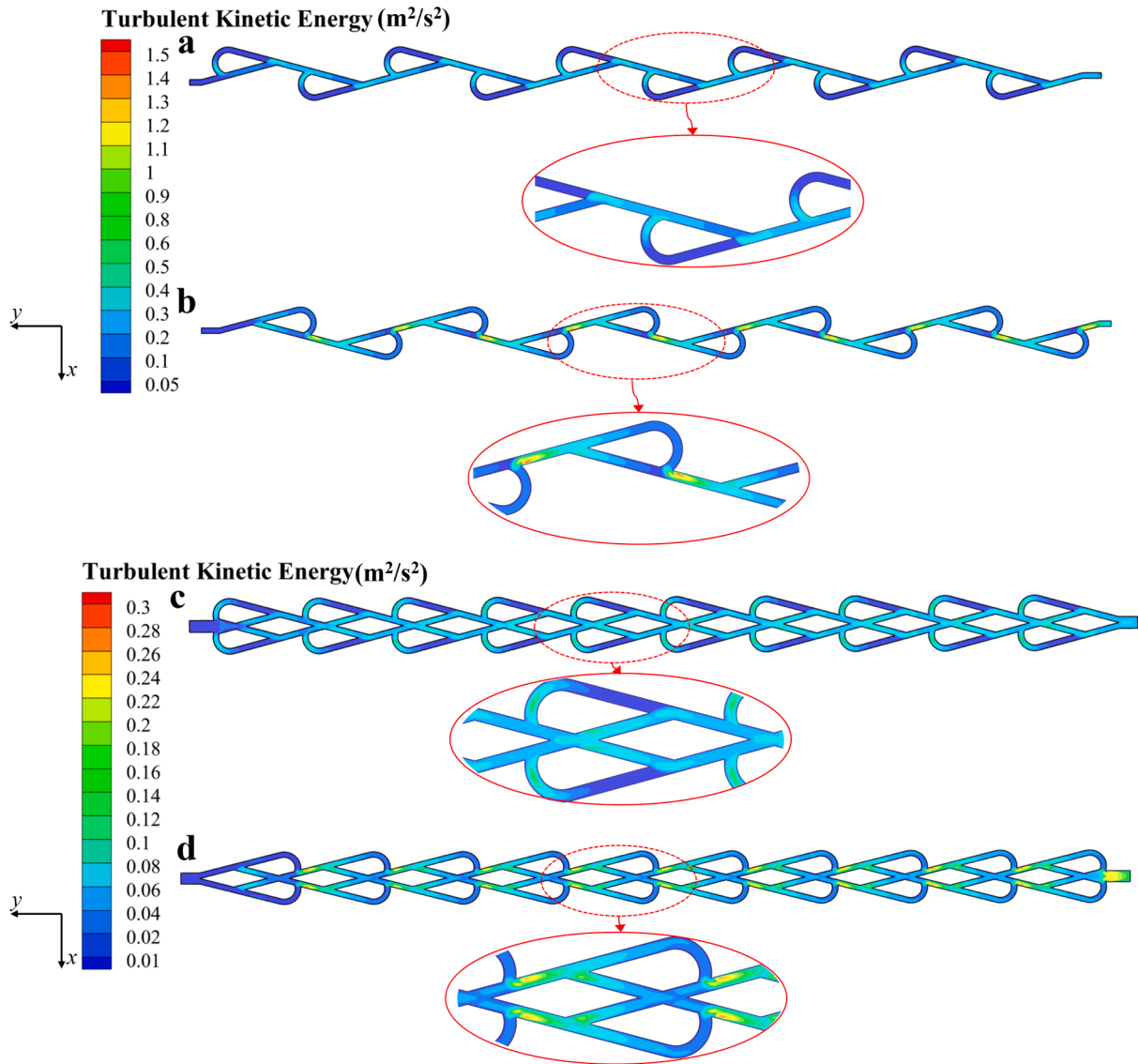


Fig. 17. Turbulent kinetic energy (*TKE*) contours at an inlet mass flow rate of 1.091 g/s: (a) TVM-Forward. (b) TVM-Reverse. (c) SYMTVM-Forward. (d) SYMTVM-Reverse.

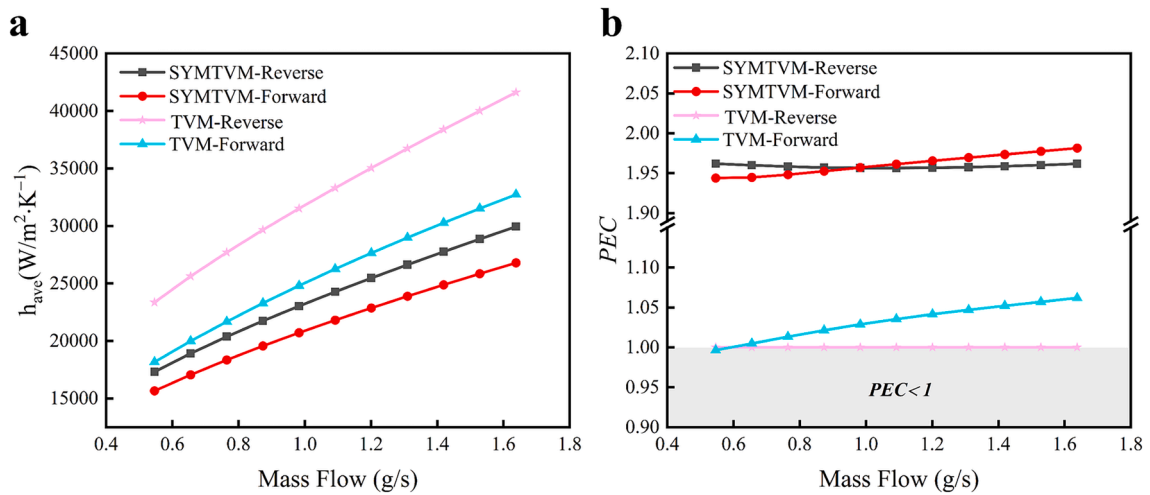


Fig. 18. (a) Changes of h_{ave} at different inlet mass flow rates. (b) Changes of PEC at different inlet mass flow rates.

the range of 0.5459 g/s to 0.98262 g/s. Above 0.98262 g/s, the PEC values for SYMTVM-Forward exhibit a stable increase, ranging from 96.1 % to 98.1 % higher than those of TVM-Reverse. Notably, PEC values of SYMTVM-Reverse maintain stable as mass flow rates increase, due to the balanced increase in Nu and pressure drop.

For the microchannel heat sinks, one essential parameter is the Maximum temperature (T_{max}). Although T_{max} is influenced by the heat transfer coefficient, the Performance Evaluation Criteria (PEC) often fails to represent T_{max} directly which generally occurs in areas with a lower heat transfer coefficient. A novel criterion known as the maximum temperature thermal resistance ($PECTR_{T_{max}}$) has been proposed in Equ. (19). This criterion explicitly includes T_{max} and ensures the pressure drop ratio aligned with PEC [47].

$$PECTR_{T_{max}} = \frac{(R_{to}/R_t)}{(\Delta P/\Delta P_0)^{1/3}} \quad (19)$$

Fig. 19 demonstrates that SYMTVM consistently outperforms TVM across all measured mass flow rates. The $PECTR_{T_{max}}$ for both the SYMTVM-Reverse and SYMTVM-Forward configurations shows a positive correlation with increasing mass flow rates, indicating enhanced heat transfer performance at higher rates. This correlation results in notably lower maximum temperatures. Specifically, the SYMTVM-Reverse configuration slightly exceeds the $PECTR_{T_{max}}$ of SYMTVM-Forward, indicating that its reverse flow arrangement dissipates heat more effectively by reducing thermal resistance. Although TVM-Forward does not surpass SYMTVM in $PECTR_{T_{max}}$, it exhibits a marginal improvement as the flow rate increases. Overall, SYMTVM-Reverse proves to be the most efficient solution, especially at higher mass flow rates.

3.4. The effect of the interconnection angle θ for trunk and helix regions in SYMTVM-Reverse

Analysis of the data in Figs. 18 and 19 leads to the conclusion that SYMTVM-Reverse is of the most efficient performance. However, the effect of the connection angle θ between the trunk and helix regions in the SYMTVM-Reverse cannot be neglected. Fig. 20 shows the structural parameters of the original SYMTVM-Reverse design with a connection angle θ of 30°. As illustrated in Fig. 21, five SYMTVM-Reverses, each with a different connection angle θ , are examined to determine how θ affects both heat transfer and flow characteristics. This study takes mass flow rates between 0.5459 g/s and 1.6377 g/s to assess the full range of operational conditions.

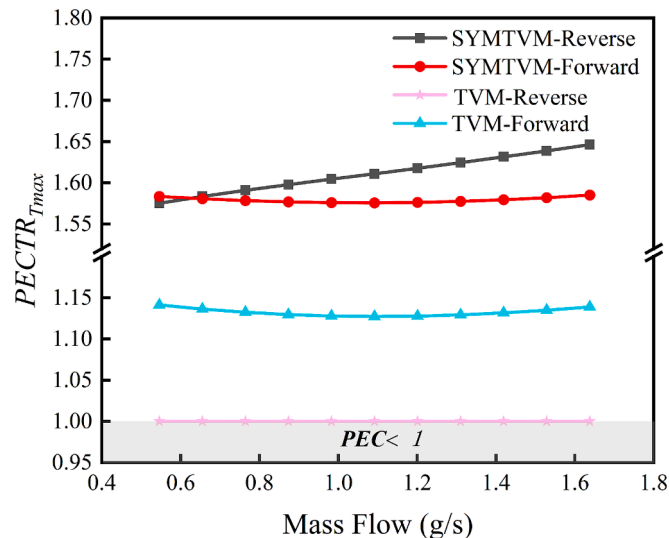


Fig. 19. Changes of $PECTR_{T_{max}}$ at varying inlet mass flow rates.

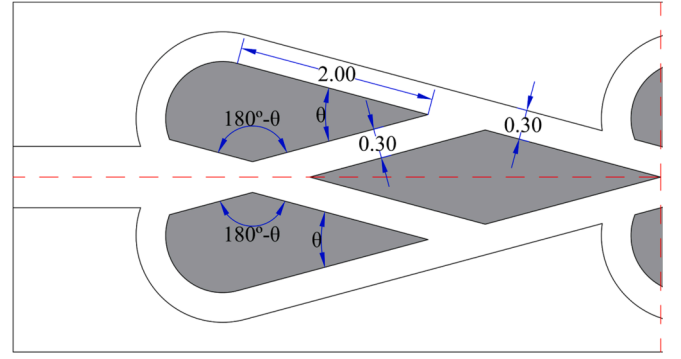


Fig. 20. Structure parameters of SYMTVM (unit: mm).

The pressure difference ΔP with the interconnection angle θ between the trunk and helix regions in SYMTVM is shown in Fig. 22a and 22b. For smaller angles ($\theta \leq 40^\circ$), there is a linear increase in ΔP with θ . However, ΔP first decreases and then increases when θ exceeds 40° , following an overall upward trajectory as θ continues to increase. As the angle increases, the length required for a single stage Tesla valve structure in SYMTVM also grows. At an angle of 50° , the effective length of the SYMTVM becomes insufficient for the converging passages typically seen at angles between 20° and 40° . This results in the formation of multiple outlets, effectively reducing pressure drop. However, a pressure drop appears at a 60° angle, which clearly demonstrates that larger angular adjustments profoundly contribute to the escalation of pressure drop. Fig. 22c and 22d, along with the provided data, reveal that the interconnection angle (θ) significantly affects the maximum temperature (T_{max}) at the heat source surface. At lower mass flow rates, such as 0.5459 g/s, T_{max} rises sharply from 377.6 K at $\theta = 20^\circ$ to 466.6 K at $\theta = 60^\circ$, indicating that larger angles reduce heat dissipation due to increased flow resistance. However, at a higher mass flow rate of 1.6377 g/s, the increase in T_{max} with θ is less pronounced, showing that enhanced convective heat transfer at higher flows mitigates the adverse effects of larger angles. This makes it of great significance to optimize θ , especially in low-flow scenarios to maintain efficient thermal performance. The relationship between the temperature difference (δT) and the interconnection angle θ for the trunk and helix regions are shown in Fig. 22e and 22f. The results indicate that decreasing angle θ effectively reduces the δT at the heat source surface of the SYMTVM-reverse. When the junction angle θ drops from 60° to 20° , a significant reduction appears in δT of 128.36 % and 127.14 % at mass flow rates of 0.5459 g/s and 1.6377 g/s, respectively.

The PEC decreases across all angles (θ) as the mass flow rate increases, with the configuration consistently achieving the highest PEC values at $\theta = 30^\circ$, as shown in Fig. 23. The superior performance at $\theta = 30^\circ$ is attributed to the optimal balance between thermal efficiency and pressure drop. At this angle, the pressure difference (ΔP) remains manageable, preventing excessive pressure losses while still promoting effective fluid mixing and heat transfer. This balance allows the configuration to maintain a higher PEC compared to other angles. In comparison, even though the pressure drop is low at smaller angles, like $\theta = 20^\circ$, the heat transfer efficiency does not reach the same optimal level as $\theta = 30^\circ$. At larger angles ($\theta \geq 50^\circ$), the increased pressure drop due to insufficient effective length of the SYMTVM structure leads to a sharp decline in PEC , particularly at $\theta = 60^\circ$, where a significant surge in pressure drop results in the lowest PEC values. Therefore, $\theta = 30^\circ$ provides the best compromise, achieving superior thermal-hydraulic performance by effectively balancing pressure drop and heat transfer efficiency.

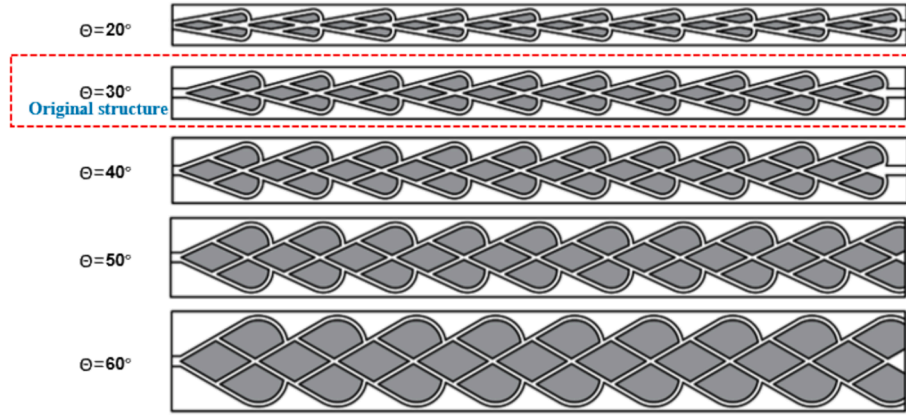


Fig. 21. Schematics of SYMTVM with different interconnection angle θ for trunk and helix regions.

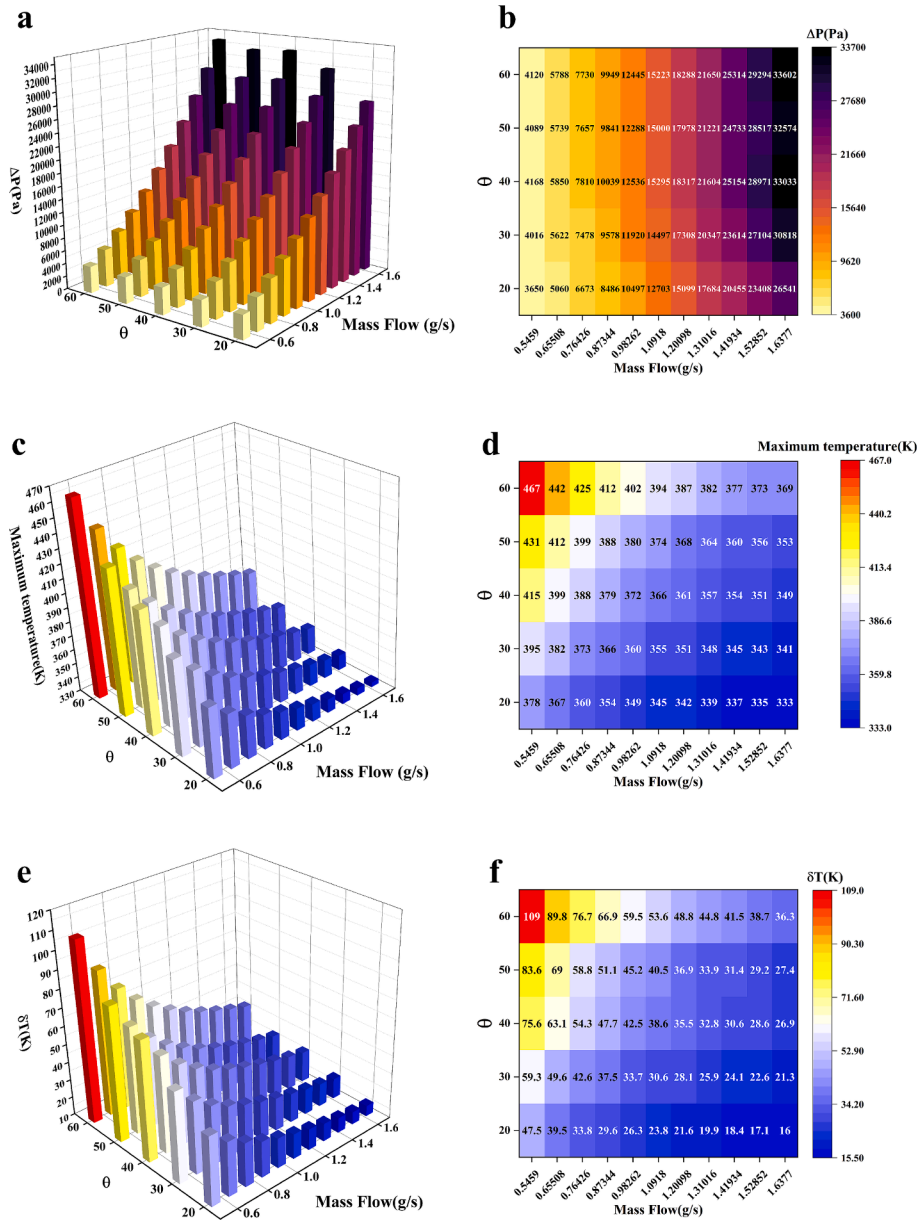


Fig. 22. Variation in performance metrics with different interconnection angles (θ) between trunk and helix regions under varying mass flow rates: (a) 3D histogram of pressure drop. (b) Heat map of pressure drop. (c) 3D histogram of maximum temperature. (d) Heat map of maximum temperature. (e) 3D histogram of temperature differences. (f) Heat map of temperature difference.

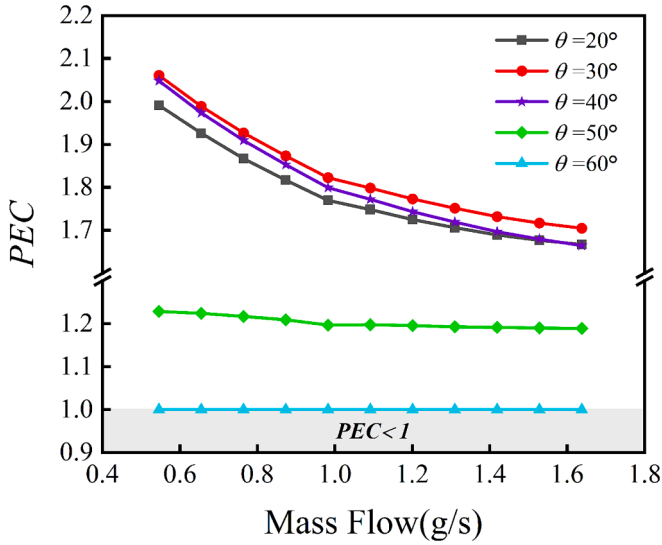


Fig. 23. Variation of PEC with different interconnection angles (θ) between trunk and helix regions under varying mass flow rates.

3.5. Comparative analysis of flow and heat transfer performance between SYMTVM-Reverse with 30° interconnection angle and rectangular microchannel (RM)

As shown in Fig. 23, the SYMTVM-Reverse with a 30°

interconnection angle demonstrates the best overall performance. To more clearly illustrate its practical significance, this section compares the SYMTVM-Reverse with a 30° interconnection angle against the traditional rectangular microchannel heat sink (RM) under different mass flow conditions, focusing on four key parameters: maximum temperature, temperature difference, Nusselt number (Nu), and power dissipation (Φ).

The comparative analysis between the SYMTVM-Reverse and RM structures under varying mass flow rates, as shown in Fig. 24, reveals that the SYMTVM-Reverse generally outperforms RM in key thermal management aspects. For maximum temperature, the SYMTVM-Reverse reduces the temperature to 340.62 K at 1.6377 g/s, which is approximately 8.5 K lower than RM's 349.17 K, indicating better heat dissipation. This improved performance is due to the reverse flow design that enhances fluid mixing, further aided by the Tesla valve's inherent resistance and the formation of lateral vortices at the trunk-helix junctions. In terms of temperature difference (δT), the SYMTVM-Reverse demonstrates a more substantial reduction compared to RM, with the temperature difference decreasing significantly across the mass flow rates. For example, at 1.6377 g/s, the temperature difference for SYMTVM-Reverse is approximately 21.26 K, whereas RM exhibits a higher difference of 33.12 K. This indicates a reduction of about 36 % in temperature difference for SYMTVM-Reverse compared to RM, leading to a more uniform temperature distribution. The significant reduction in δT is primarily due to the continuous disruption of the thermal boundary layer within the SYMTVM-Reverse, facilitated by its complex flow dynamics, which promote sustained vortex generation and mixing. Additionally, the Nusselt number (Nu) for SYMTVM-Reverse reaches 41.71 at 1.6377 g/s, significantly higher than RM's 8.76, reflecting its more

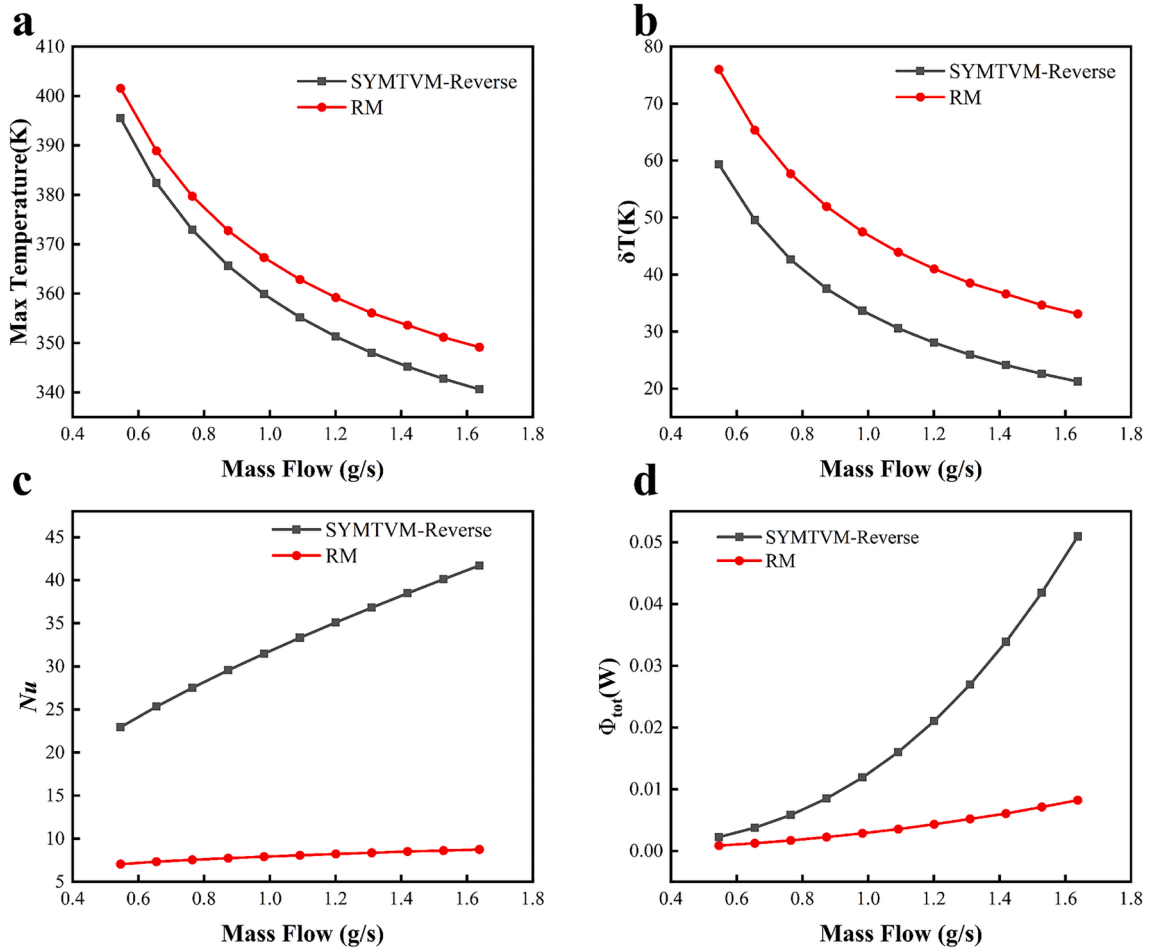


Fig. 24. Variations in (a) T_{max} , (b) δT , (c) Nu and (d) Φ_{tot} for SYMTVM-Reverse and RM under varying inlet mass flow rates.

efficient convective heat transfer. However, the enhanced thermal performance comes at the cost of higher power dissipation, with the SYMTVM-Reverse consuming about 0.05093 W at 1.6377 g/s, compared to RM's lower consumption of 0.00822 W, highlighting the trade-off between thermal efficiency and energy consumption.

To accurately assess the performance between the SYMTVM-Reverse and RM structures, it is important to consider the significant difference in inlet mass flow between a single SYMTVM-Reverse unit and a RM computational unit, with the former being six times greater than the latter. The pressure drop component in the standard PEC formula does not adequately reflect this discrepancy in comprehensive performance. Therefore, to facilitate a more precise quantitative comparison of the overall performance between the SYMTVM-Reverse and RM structures, we introduced a modified version of the PEC formula, termed $PEC_{\Phi_{tot}}$. In this modified formula, the pressure drop term is replaced by the total power dissipation of a single SYMTVM-Reverse unit and the combined power dissipation of six RM channels. The $PEC_{\Phi_{tot}}$ formula is expressed as follows:

$$PEC_{\Phi_{tot}} = \frac{(Nu/Nu_0)}{(\Phi_{tot}/\Phi_{tot0})^{1/3}} \quad (20)$$

Fig. 25 illustrates that the $PEC_{\Phi_{tot}}$ values for SYMTVM-Reverse range from approximately 2.37 to 2.59, indicating that the overall performance of the SYMTVM-Reverse structure is 2.37 to 2.59 times better than that of the RM structure across the tested mass flow rates, which shows the superiority of SYMTVM-Reverse in balancing heat transfer efficiency with pressure drop, clearly outperforming the RM structure which serves as the baseline ($PEC_{\Phi_{tot}} = 1$).

4. Conclusions

In this research, the performance of symmetrical Tesla-type microchannels were investigated, aiming to enhance the efficiency of multi-stage Tesla microchannel heat sinks. Two Tesla-type microchannels were evaluated under forward and reverse flow conditions, focusing on their heat transfer and hydraulic characteristics. The main conclusions are:

- (1) In reverse flow, the TVM and SYMTVM exhibited superior heat transfer performance compared to forward flow. Specifically, the maximum temperature was significantly reduced, and temperature uniformity was enhanced by 26.4 % and 26.27 % for TVM and SYMTVM, respectively.
- (2) Reverse flow resulted in the highest TVM pressure drop, up to 5.87 times higher than forward flow conditions, and increased turbulent kinetic energy (TKE), especially in the divergence and convergence regions of the Tesla valve cells to enhance heat transfer.
- (3) Localized vortices at the trunk/helix outlets in TVM-Forward and SYMTVM-Forward and complex flow patterns in TVM-Reverse and SYMTVM-Reverse led to unstable multidirectional flow and improved heat transfer.
- (4) Under reverse flow, SYMTVM showed the best thermo-hydraulic performance with PEC values between 1.957 and 1.961 and $PECTR_{Tmax}$ between 1.575 and 1.946 as more vortices and bifurcations disrupted and redeveloped the thermal boundary layer. In addition, a smaller connection angle θ between the trunk and helix regions improved the heat transfer performance.
- (5) The SYMTVM-Reverse with a 30° interconnection angle significantly outperformed the RM structure, achieving a Nusselt number (Nu) nearly five times higher and an overall performance ($PEC_{\Phi_{tot}}$) up to 2.59 times better, demonstrating its superior heat transfer efficiency and balanced thermo-hydraulic performance.

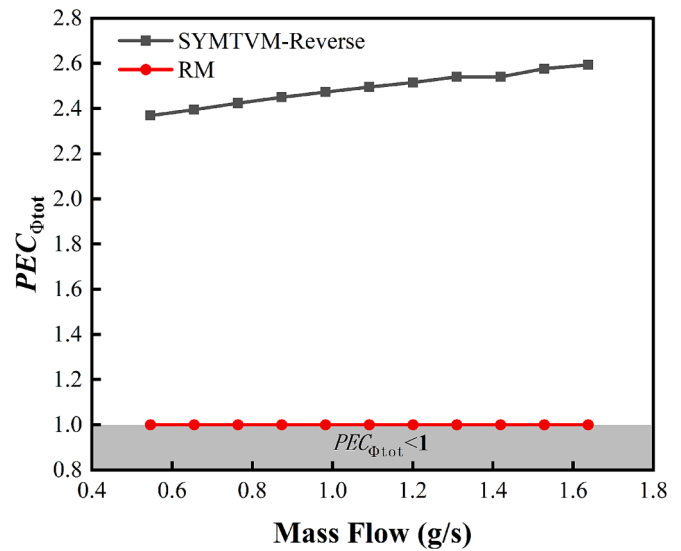


Fig. 25. Comparison of Overall Performance Efficiency ($PEC_{\Phi_{tot}}$) between SYMTVM-Reverse with 30° interconnection angle and RM structures.

CRediT authorship contribution statement

Yongqi Xia: Visualization, Investigation, Writing – original draft, Validation, Data curation. **Mingtao Wu:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Shibo Deng:** Conceptualization, Methodology, Software, Investigation, Formal analysis. **Gaozhan Yuan:** Investigation, Resources. **Quanli Zhang:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] I. Mudawar, Recent advances in high-flux, two-phase thermal management, J. Therm. Sci. Eng. Appl. 5 (2) (2013) 021012, <https://doi.org/10.1115/1.4023599>.
- [2] M. Pedram, S. Nazarian, Thermal modeling, analysis, and management in VLSI circuits: principles and methods, Proc. IEEE 94 (8) (2006) 1487–1501, <https://doi.org/10.1109/jproc.2006.879797>.
- [3] D.B. Tuckerman, R.F.W. Pease, High-performance heat sinking for VLSI, IEEE Electron Device Lett. EDL-2(5) (1981) 126–129, <https://doi.org/10.1109/edl.1981.25367>.
- [4] S.G. Kandlikar, W.J. Grande, Evolution of microchannel flow passages—thermo-hydraulic performance and fabrication technology, Heat Transf. Eng. 24 (3–4) (2003) 3–17, <https://doi.org/10.1080/01457630304040>.
- [5] M.U. Sajid, H.M. Ali, Recent advances in the application of nanofluids in heat transfer devices: a critical review, Renew. Sustain. Energy Rev. 103 (2019) 556–592, <https://doi.org/10.1016/j.rser.2018.12.057>.

- [6] M.U. Sajid, Y. Bicer, Experimental investigation of spectrum selective nanofluids for thermal management of greenhouses in hot arid climates, *Appl. Therm. Eng.* 223 (2023) 120019, <https://doi.org/10.1016/j.applthermaleng.2023.120019>.
- [7] M.U. Sajid, H.M. Ali, Y. Bicer, Exergetic performance assessment of magnesium oxide–water nanofluid in corrugated minichannel heat sinks: an experimental study, *Int. J. Energy Res.* 46 (8) (2022) 9985–10001, <https://doi.org/10.1002/er.60246>.
- [8] A.F. Al-Neama, Z. Khatir, N. Kapur, J. Summers, H.M. Thompson, An experimental and numerical investigation of chevron fin structures in serpentine minichannel heat sinks, *Int. J. Heat Mass Transf.* 120 (2018) 1213–1228, <https://doi.org/10.1016/j.ijheatmasstransfer.2017.12.092>.
- [9] W. Duangthongsuk, S. Wongwises, An experimental investigation on the heat transfer and pressure drop characteristics of nanofluid flowing in microchannel heat sink with multiple zigzag flow channel structures, *Exp. Therm Fluid Sci.* 87 (2017) 30–39, <https://doi.org/10.1016/j.expthermflusci.2017.04.013>.
- [10] A.Y.M. Ali, E.M. Abo-Zahhad, H.I. Elqady, M. Rabie, M.F. Elkady, S. Ookawara, A. H. El-Shazly, A. Radwan, Thermal analysis of high concentrator photovoltaic module using convergent-divergent microchannel heat sink design, *Appl. Therm. Eng.* 183 (2021) 116201, <https://doi.org/10.1016/j.applthermaleng.2020.116201>.
- [11] S. Chamoli, R. Lu, H. Chen, Y. Cheng, P. Yu, Numerical optimization of design parameters for a modified double-layer microchannel heat sink, *Int. J. Heat Mass Transf.* 138 (2019) 373–389, <https://doi.org/10.1016/j.icheatmasstransfer.2010.09.012>.
- [12] R.S. Andhare, A. Shooshtari, S.V. Dessiatoun, M.M. Ohadi, Heat transfer and pressure drop characteristics of a flat plate manifold microchannel heat exchanger in counter flow configuration, *Appl. Therm. Eng.* 96 (2016) 178–189, <https://doi.org/10.1016/j.applthermaleng.2015.10.133>.
- [13] L. Pan, Z. Yao, W. Yao, X. Wei, Multi-objective optimization on bionic fractal structure for heat exchanging of two fluids by genetic algorithm, *Int. J. Heat Mass Transf.* 212 (2023) 124298, <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124298>.
- [14] H.A. Mohammed, P. Gunasegaram, N.H. Shuaib, Numerical simulation of heat transfer enhancement in wavy microchannel heat sink, *Int. Commun. Heat Mass Transf.* 38 (1) (2011) 63–68, <https://doi.org/10.1016/j.icheatmasstransfer.2010.09.012>.
- [15] M. Rahimi, M. Hajialyani, R. Beigzadeh, A.A. Alsairafi, Application of artificial neural network and genetic algorithm approaches for prediction of flow characteristic in serpentine microchannels, *Chem. Eng. Res. Des.* 98 (2015) 147–156, <https://doi.org/10.1016/j.cherd.2015.05.005>.
- [16] X. Zhou, C. Zeng, Y. Song, M. Jiao, F. Zhang, M. Liu, Experimental study on heat transfer and flow resistance performance of a microchannel heat exchanger with zigzag flow channels, *Prog. Nucl. Energy* 147 (2022) 104190, <https://doi.org/10.1016/j.pnucene.2022.104190>.
- [17] L. Lin, Y.-Y. Chen, X.-X. Zhang, X.-D. Wang, Optimization of geometry and flow rate distribution for double-layer microchannel heat sink, *Int. J. Therm. Sci.* 78 (2014) 158–168, <https://doi.org/10.1016/j.ijthermalsci.2013.12.009>.
- [18] K. Tang, G. Lin, Y. Guo, J. Huang, H. Zhang, J. Miao, Simulation and optimization of thermal performance in diverging/converging manifold microchannel heat sink, *Int. J. Heat Mass Transf.* 200 (2023) 123495, <https://doi.org/10.1016/j.ijheatmasstransfer.2022.123495>.
- [19] J. Zhou, Q. Li, X. Chen, Micro pin fins with topologically optimized configurations enhance flow boiling heat transfer in manifold microchannel heat sinks, *Int. J. Heat Mass Transf.* 206 (2023) 123495, <https://doi.org/10.1016/j.ijheatmasstransfer.2023.123495>.
- [20] Y. Yan, H. Yan, S. Yin, L. Zhang, L. Li, Single/multi-objective optimizations on hydraulic and thermal management in micro-channel heat sink with bionic Y-shaped fractal network by genetic algorithm coupled with numerical simulation, *Int. J. Heat Mass Transf.* 129 (2019) 468–479, <https://doi.org/10.1016/j.ijheatmasstransfer.2018.09.120>.
- [21] Y. Sui, C.J. Teo, P.S. Lee, Y.T. Chew, C. Shu, Fluid flow and heat transfer in wavy microchannels, *Int. J. Heat Mass Transf.* 53 (13–14) (2010) 2760–2772, <https://doi.org/10.1016/j.ijheatmasstransfer.2010.02.022>.
- [22] G. Lu, X. Zhai, Analysis on heat transfer and pressure drop of a microchannel heat sink with dimples and vortex generators, *Int. J. Therm. Sci.* 145 (2019) 105986, <https://doi.org/10.1016/j.ijthermalsci.2019.105986>.
- [23] G. Xia, L. Chai, H. Wang, M. Zhou, Z. Cui, Optimum thermal design of microchannel heat sink with triangular reentrant cavities, *Appl. Therm. Eng.* 31 (6–7) (2011) 1208–1219, <https://doi.org/10.1016/j.applthermaleng.2010.12.022>.
- [24] P. Bhandari, Y.K. Prajapati, Thermal performance of open microchannel heat sink with variable pin fin height, *Int. J. Therm. Sci.* 159 (2021) 106609, <https://doi.org/10.1016/j.ijthermalsci.2020.106609>.
- [25] C. Chen, F. Li, X. Wang, J. Zhang, G. Xin, Improvement of flow and heat transfer performance of manifold microchannel with porous fins, *Appl. Therm. Eng.* 206 (2022) 118129, <https://doi.org/10.1016/j.applthermaleng.2022.118129>.
- [26] Z. Wu, B. Sunden, On further enhancement of single-phase and flow boiling heat transfer in micro/minichannels, *Renew. Sustain. Energy Rev.* 40 (2014) 11–27, <https://doi.org/10.1016/j.rser.2014.07.171>.
- [27] J.-F. Zhang, L. Jia, W.-W. Yang, J. Taler, P. Oclon, Numerical analysis and parametric optimization on flow and heat transfer of a microchannel with longitudinal vortex generators, *Int. J. Therm. Sci.* 141 (2019) 211–221, <https://doi.org/10.1016/j.ijthermalsci.2019.03.036>.
- [28] E. Khodabandeh, S.A. Rozati, M. Joushaghani, O.A. Akbari, S. Akbari, D. Toghraie, Thermal performance improvement in water nanofluid/GNP-SDBS in novel design of double-layer microchannel heat sink with sinusoidal cavities and rectangular ribs, *J. Therm. Anal. Calorim.* 136 (3) (2019) 1333–1345, <https://doi.org/10.1007/s10973-018-7826-2>.
- [29] L. Yin, M. Sun, P. Jiang, C. Dang, L. Jia, Heat transfer coefficient and pressure drop of water flow boiling in porous open microchannels heat sink, *Appl. Therm. Eng.* 218 (2023) 119361, <https://doi.org/10.1016/j.applthermaleng.2022.119361>.
- [30] L. Sun, J. Li, H. Xu, J. Ma, H. Peng, Numerical study on heat transfer and flow characteristics of novel microchannel heat sinks, *Int. J. Therm. Sci.* 176 (2022) 107535, <https://doi.org/10.1016/j.ijthermalsci.2022.107535>.
- [31] Z. Rui, F. Zhao, H. Sun, L. Sun, H. Peng, Experimental research on flow boiling thermal-hydraulic characteristics in novel microchannels, *Exp. Therm Fluid Sci.* 140 (2023), <https://doi.org/10.1016/j.expthermflusci.2022.110755>.
- [32] X. Li, X. Zhang, W. Li, Z. Wang, X. Zhang, Y. Zhao, The 3D Tesla Valve Manifold Two-Phase Cold Plate for the SiC Power Device Module, In: Proceedings of the 2022 21st IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm), San Diego, CA, USA, 2022, pp. 1–8. <https://doi.org/10.1109/ITherm54085.2022.9899661>.
- [33] J. Qian, M. Chen, X. Li, Z. Jin, Numerical investigation of heat sink with tesla valve type microchannel, *Fluid Mach.* 48 (12) (2020) 36–42.
- [34] P.R. Porwal, S.M. Thompson, D.K. Walters, T. Jamal, Heat transfer and fluid flow characteristics in multistaged Tesla valves, *Numer. Heat Transf. Part A-Appl.* 73 (6) (2018) 347–365, <https://doi.org/10.1080/10407782.2018.1447199>.
- [35] F. Gong, X. Yang, X. Zhang, Z. Mao, W. Gao, C. Wang, The study of Tesla valve flow field on the net power of proton exchange membrane fuel cell, *Appl. Energy* 329 (2023) 120276, <https://doi.org/10.1016/j.apenergy.2022.120276>.
- [36] Q. Han, Z. Liu, Y. Chen, W. Li, Enhance flow boiling in Tesla-type microchannels by inhibiting two-phase backflow, *Int. J. Heat Mass Transf.* 214 (2023) 124471, <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124471>.
- [37] N. Tesla, Valvular Conduit, U.S. Patent No. 1,329,559, 1920.
- [38] Q. Zhu, R. Su, L. Hu, J. Chen, J. Zeng, H. Zhang, H. Sun, S. Zhang, D. Fu, Heat transfer enhancement for microchannel heat sink by strengthening fluids mixing with backward right-angled trapezoidal grooves in channel sidewalls, *Int. Commun. Heat Mass Transf.* 135 (2022) 106106, <https://doi.org/10.1016/j.icheatmasstransfer.2022.106106>.
- [39] K. Derakhshanpour, R. Kamali, M. Eslami, Improving performance of single and double-layered microchannel heat sinks by cylindrical ribs: a numerical investigation of geometric parameters, *Int. Commun. Heat Mass Transf.* 126 (2021) 105440, <https://doi.org/10.1016/j.icheatmasstransfer.2021.105440>.
- [40] G. Wang, T. Chen, M. Tian, G. Ding, Fluid and heat transfer characteristics of microchannel heat sink with truncated rib on sidewall, *Int. J. Heat Mass Transf.* 148 (2020) 119142, <https://doi.org/10.1016/j.ijheatmasstransfer.2019.119142>.
- [41] Q.M. Nguyen, J. Abouezzi, L. Ristoph, Early turbulence and pulsatile flows enhanceodicity of Tesla's macrofluidic valve, *Nat. Commun.* 12 (1) (2021) 2884, <https://doi.org/10.1038/s41467-021-23009-y>.
- [42] X. Rao, C. Jin, H. Zhang, J. Song, C. Xiao, A hybrid microchannel heat sink with ultra-low pressure drop for hotspot thermal management, *Int. J. Heat Mass Transf.* 211 (2023) 124201, <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124201>.
- [43] A.R. Gamboa, C.J. Morris, F.K. Forster, Improvements in fixed-valve micropump performance through shape optimization of valves, *J. Fluids Eng.* 127 (2) (2005) 339–346, <https://doi.org/10.1115/1.1891151>.
- [44] Y. Bao, H. Wang, Numerical study on flow and heat transfer characteristics of a novel Tesla valve with improved evaluation method, *Int. J. Heat Mass Transf.* 187 (2022) 122540, <https://doi.org/10.1016/j.ijheatmasstransfer.2022.122540>.
- [45] F.P. Incropera, Convection heat transfer in electronic equipment cooling, *J. Heat Transfer* 110 (4b) (1988) 1097–1111, <https://doi.org/10.1115/1.3250613>.
- [46] C.J. Ho, L.C. Wei, Z.W. Li, An experimental investigation of forced convective cooling performance of a microchannel heat sink with Al₂O₃/water nanofluid, *Appl. Therm. Eng.* 30 (1) (2010) 96–103, <https://doi.org/10.1016/j.applthermaleng.2009.07.003>.
- [47] J. Zhang, J. An, G. Xin, X. Wang, J. Huang, L. Li, Z. Wu, Thermal and hydrodynamic characteristics of single-phase flow in manifold microchannels with countercurrent regions, *Int. J. Heat Mass Transf.* 211 (2023) 124265, <https://doi.org/10.1016/j.ijheatmasstransfer.2023.124265>.